Intelligent Auditing of Internet Infrastructure

Bounded Claims from Partial Evidence

PhD prospectus

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Abstract

Internet services expose addresses, names, routing identifiers, and response endpoints. Operators and users need to know something different: where the infrastructure operates, which organizations perform particular roles, and who participates in handling a request. A hostname may suggest several cities, several ASNs may represent one operator, and a selected DNS resolver may rely on unreported intermediaries. These gaps complicate location-based decisions, concentration analysis, service attribution, and dependency assessment.

This dissertation argues that partial evidence can support bounded conclusions about infrastructure location, operational identity, and request participation when the interpretation from observation to claim is explicit and the resulting claim is tested against appropriate evidence. Where existing observations cannot answer the question, cooperation with infrastructure participants can extend what can be known.

The location research begins with The Aleph, which uses language models to learn reusable regular expressions and operator-specific geographic mappings from a sample of PTR records. A diverse 30-AS benchmark exposes its remaining recovery gap and motivates Bet's structure-aware sampling and explicit mapping of extracted vocabulary. On 506 held-out hostnames with geographic references, mean recall of annotated geographic hints rises from 70.6% to 90.9%. Operator review and RTT constraints separately examine semantic meaning and physical compatibility. Motivated by customer complaints about wrongly blocked or improperly allowed access, GeoCorrect asks what minimum location evidence is needed to support or challenge a geographic access decision, even when the exact location remains unknown.

Studies of network concentration and service dependencies must distinguish two operational-identity errors: *false independence*, in which several ASNs make one operator appear to be several, and *false unity*, in which one network identity conceals multiple service actors. Miscounting examines false independence by comparing concentration and change under dated ASN-to-operator mappings. Preliminary sensitivity and churn findings motivate the remaining mapping audit and longitudinal analysis. GOVT-MX examines false unity in government email by distinguishing the providers responsible for inbound handling, hosting, and authorized sending.

Finally, DNSTRACING asks who participates in handling a DNS request. Observed endpoints alone cannot answer this question: different sequences of intermediaries can produce the same visible endpoints. The research examines this visibility gap and develops cooperative reporting to make otherwise hidden participation inspectable. Authentication verifies the origin and response binding of received reports; it does not guarantee complete participation or the accuracy of reported assertions. Controlled evaluation establishes feasibility.

Remaining empirical work comprises GeoCorrect's full-day and month-long analyses, Miscounting's mapping and longitudinal validation, and Resolver-Path's option-survival, key-lifecycle, and deployment tests. The dissertation will provide inspectable methods and artifacts for determining what incomplete observations support and what remains unresolved.

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Chapter 1

Introduction

Internet observations reveal only part of the infrastructure behind a service. This chapter introduces the location, operator-identity, and request-participation questions that organize the dissertation, and explains what kinds of answers partial evidence can support.

1.1 Where does the infrastructure operate?

A video provider buys a delivery service and configures its application in a known hosting region. It can identify the addresses serving its viewers, yet that does not establish where the delivery infrastructure operates. Different viewers may reach different facilities, and the application's configured region need not locate every delivery endpoint. To understand the service's footprint, the provider needs to ask: *where are these endpoints?*

A reverse-DNS lookup supplies a clue: `edge.mad.example.net`. In this illustrative hostname, the token `mad` could suggest Madrid, but an operator might also use it for Madison. The name gives a candidate interpretation rather than a self-explanatory coordinate. The first question is therefore how to interpret the name; the next is what independent evidence could distinguish the candidate locations. We return to this example after considering why the distinction matters.

Location records can directly affect an operational decision. Suppose a payment API blocks requests labeled as coming from country B. An address-to-location record labels a merchant's connection B, so the API rejects the request. The record might come from a commercial geolocation database or an operator-published geofeed [1]. These sources describe where the label came from; they do not independently confirm its physical meaning. If the merchant disputes the label, *what evidence can test whether the country restriction should apply?*

Maintaining such information is also a scaling problem. IPv4's 32-bit format admits about 4.29 billion addresses, while IPv6's 128-bit format admits about 3.4×10^{38} [2]. Assignments and naming conventions can change over time. Useful methods must therefore recover and reuse information without assuming that every address can be individually interpreted or independently located.

1.2 Who operates the infrastructure?

A bank assessing a critical payment service begins with contracts and its service register. Those records identify declared providers and services; network observations can help investigate the infrastructure supporting them. For example, a current connection and an alternative connection may traverse two differently named networks. If both ASNs belong to one operator, counting them as independent suppliers overstates diversity. If they are unrelated, merging them conceals diversity. *Do two network identifiers represent two operators?* Registration and other public records can inform an answer, but a routing identifier alone cannot supply it.

This question matters to resilience planning and to regulatory assessments of provider concentration. DORA provides one concrete motivation for examining service locations, providers, and third-party dependencies [3]. The research question here is how concentration estimates change under explicit interpretations of network identity.

Government email exposes the opposite difficulty. A public-sector domain looks like one service, and its mail endpoint may sit inside one hosting ASN. Yet a gateway may handle inbound messages, another organization may host the machines, and a third may be authorized to send messages for the domain. An MX record, application greeting, hosting network, and SPF authorization answer different parts of the question. *Who performs each role?* Treating the hosting ASN as the email operator loses distinctions that matter to an account of operational responsibility. An authorization to send, in particular, is not evidence that a message was sent.

1.3 Who else handled the request?

A user chooses a DNS resolver partly because of its privacy policy. The client can observe the address it contacts; an authoritative server can observe the source of a query that reaches it. The selected resolver may nevertheless forward queries through other infrastructure. Even when observations identify those two endpoints, more than one hidden resolver arrangement can explain them. *Who else handled this query?*

The question differs from an ambiguous hostname or an uncertain organization mapping. Ordinary DNS observations do not report every intervening resolver. Endpoint measurements can establish endpoint behavior, but they cannot recover a participant that leaves no distinguishing observation. Learning about that participant requires a cooperative disclosure mechanism. Checking the origin and response binding of the resulting report is a separate task from obtaining it, and neither task proves that every participant disclosed itself.

1.4 What can partial evidence establish?

Return to the Madrid/Madison hostname. RTT observations from known vantage points can constrain the endpoint's possible distance from those vantage points under a propagation model. Suppose the resulting feasible region contains Madison and excludes Madrid, as illustrated in Figure 1.1. The appropriate conclusion is that *Madison remains physically compatible; the exact site remains unknown.* The delay evidence does not independently confirm that the operator intended `mad` to mean Madison, nor does it select one facility within the feasible region.

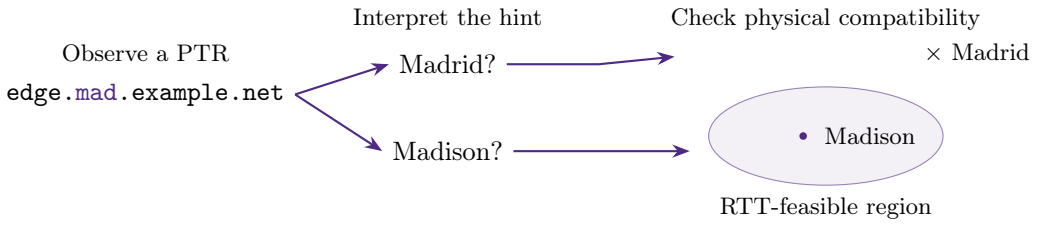


Figure 1.1: Illustrative hostname ambiguity and a conditional RTT check. The region is schematic, not a map or measured result. Compatibility retains a location hypothesis; it does not confirm hostname meaning or an exact site.

The other examples require the same discipline with different evidence. Alternative ASN groupings can expose the sensitivity of a concentration estimate. Mail records can support distinct operational roles. Cooperative DNS reports can add information that endpoint observations leave unreported. The useful conclusion is the one the available evidence supports, including its unresolved alternatives.

Thesis statement. Partial evidence can support bounded conclusions about infrastructure location, operational identity, and request participation when the interpretation from observation to claim is explicit and the resulting claim is tested against appropriate evidence.

When existing observations cannot answer the question, the dissertation also examines how cooperative disclosure can produce new evidence and how attestation can authenticate received reports. This extends the available evidence without assuming complete participation.

1.5 Research questions and bounded scope

RQ1: *How can names yield scalable candidate locations, and when can independent measurements reject or support a consequential location claim?* Aleph and Bet recover geographic hints; GeoCorrect tests disputed location records using passive RTT evidence.

RQ2: *How do estimates of longitudinal concentration change under explicit ASN-to-operator identity assumptions?* Miscounting compares dated mappings over routing observations and evaluates their consequences.

RQ3: *How can observations distinguish the organizations performing different service roles?* GOVT-MX separates inbound handling, hosting, and authorized sending through role-specific evidence.

RQ4: *What can cooperating resolvers disclose and authenticate about query participation when ordinary observations omit intermediaries?* DNSTRACING connects endpoint measurement to cooperative reporting and response-bound attestation.

Table 1.1: Bounding the thesis scope. Each supported property has an adjacent limit.

Property	In scope	Limit
Location	Candidate locations, physical compatibility, exclusions and supported corrections	No exact universal placement, data residency, or jurisdiction
Operator grouping	Concentration and change under explicit identity models	No complete operator census or independent-failure guarantee
Email roles	Inbound handling, hosting, and sending authorization	No mailbox residency or inference of sending from SPF alone
DNS participation	Endpoint observations and authenticated cooperative reports	No complete hidden path or proof that silent intermediaries are absent

1.6 Dissertation roadmap

Chapters 2–3 establish the related work and the evidence distinctions used throughout the dissertation. Part I follows the location research through The Aleph, Bet, and GeoCorrect: from interpreting hostname hints to testing consequential location claims. Part II examines operator grouping in Miscounting and service-role attribution in GOVT-MX. Part III begins with DNS endpoint measurement and then develops cooperative disclosure and attestation for otherwise unreported participation. Chapters 11–14 bring the research parts together in the remaining empirical work, evaluation criteria, contributions, and target milestones. The examples above motivate those studies.

Chapter 2

Background and related work

This dissertation contributes methods for recovering and interpreting geographic hints, testing location claims, distinguishing operators and service roles, and obtaining authenticated DNS resolver reports. This chapter explains the operational and regulatory questions that motivate these contributions and relates them to prior Internet measurement research.

2.1 Regulatory and operational requirements for service visibility

The dissertation's methods provide evidence about where infrastructure operates, which organizations perform particular service roles, and which cooperating resolvers report handling a request. These capabilities support operational decisions and offer technical checks on declared service dependencies. Regulatory records identify declared providers and obligations; operational telemetry records behavior in systems that an organization operates, instruments, or contracts.

DORA provides a concrete regulatory motivation in the European financial sector [3]. Article 8 requires financial entities to identify ICT-supported functions, assets, dependencies, and third-party interconnections. Article 29 requires an entity-level assessment before contracting a provider that is difficult to substitute or entering multiple arrangements with the same or closely connected providers. Article 30 requires contracts to specify services, subcontracting conditions, and the countries where contracted or subcontracted functions and services are provided and data are processed or stored.

To apply the dissertation's methods in such an audit, an auditor would link a declared ICT service and its register entry to network endpoints observed at the relevant time. Location checks, operator groupings, and resolver reports could then identify discrepancies for the financial entity or competent authority to investigate. Appendix A maps the detailed register fields to these technical checks and identifies the contractual information each check requires.

Operational guidance calls for related visibility for different reasons. AWS recommends telemetry for dependencies including DNS, network routes, and external services [4]. ThousandEyes documents network- and application-layer testing and path visualization for diagnosing service disruptions [5], while Cloudflare documents device, network, and application performance measurement [6]. Cloudflare's load-balancing architecture further uses health, geography, proximity, and round-trip time to select and fail over between endpoints [7]. For applications under common instrumentation, OpenTelemetry propagates context across participating services so that a tracing system can join telemetry into a request trace [8].

These technical checks complement the internal telemetry and contractual records used in operational and regulatory assessments. They do not alone establish legal ownership, data residency, register completeness, regulatory compliance, or service resilience.

2.2 Measuring service-facing Internet infrastructure

Internet measurement observes protocol objects rather than service-level properties. A domain name may resolve through several DNS providers, return an anycast address, terminate at a shared gateway, and depend on additional providers that never appear in a client-visible response. The

translation from those objects to a location, actor, service role, or participant set is therefore a measurement decision rather than a neutral relabeling step.

The Border Gateway Protocol (BGP) exposes advertised prefixes and origin ASNs. DNS exposes records and visible query endpoints. Traceroute exposes a vantage-dependent sequence of interfaces. Application handshakes expose only the role and state implemented at the contacted endpoint. Translating any of these observations into a claim about a service therefore requires an explicit unit of analysis.

Prior measurement systems illustrate the complementary evidence available at these layers. Rocketfuel combined traceroutes, routing data, and DNS names to infer Internet service provider (ISP) topologies [9]. OpenINTEL performs large-scale active DNS measurement, providing longitudinal protocol observations rather than a direct inventory of service operators [10]. Client-side DNS measurements show that stub configurations, local forwarders, and recursive infrastructures form a more complex population than a configured resolver address alone suggests [11]. Liu et al. demonstrate a sharper endpoint ambiguity: on-path devices can intercept a query addressed to a selected resolver and return an answer under the selected resolver's source address [12].

These techniques also expose recurring methodological limits. Active results depend on vantage-point placement and observation time. Passive routing data describes control-plane visibility rather than every request's application path. Caching, load balancing, forwarding, and anycast complicate DNS and application measurements. Datasets collected at different times can also produce invalid joins.

An observed IP address can accordingly support a statement about a response endpoint without identifying every dependency behind it. An origin ASN can support a routing statement without identifying an independent provider. A server greeting can identify software or a presented service name without establishing legal ownership.

The dissertation treats this translation as part of the measurement rather than as a data-cleaning detail. Aleph and Bet attach location interpretations to the PTR metadata and operator context that informed those interpretations. Miscounting tests whether conclusions survive alternative mappings from routing identifiers to organizations. GOVT-MX orders network, application, and authorization signals according to the mail role each describes. DNSTRACING separates the client-facing and authoritative-facing resolver endpoints and introduces cooperative reporting for participation that endpoint observation cannot recover. Across the studies, coverage, provenance, observation time, translation rule, and supported claim remain coupled.

2.3 Infrastructure location and traffic steering

IP geolocation methods draw on several evidence classes. Early systems used network delay, topology, and external geographic landmarks to constrain a host's possible location [13]. Commercial and public databases provide convenient address-to-location mappings, but their accuracy varies by address type and by the data available to the provider. Self-published geofeeds instead allow a network operator to publish coarse prefix-to-location mappings in a common format [1]. A geofeed is therefore a direct operator statement for its covered prefixes, although publication coverage, freshness, and prefix granularity still limit the claims a consumer can make.

Reverse DNS supplies a different source: operators often encode geographic hints in interface and infrastructure names for troubleshooting. DRoP derives domain-specific rules from geographic dictionaries and validates candidate locations with topology and performance evidence [14]. HLOC similarly extracts hostname hints and tests them using measurements from multiple frameworks [15]. Hoiho learns regular expressions for operator naming domains, uses delay constraints, and incorporates

operator feedback [16].

Dan et al. broaden reverse-DNS geolocation beyond router-only datasets, combining hostname segmentation and feature-based inference [17]. Together, these systems establish that PTR records are useful at scale, while also showing why a global dictionary is insufficient: abbreviations collide, naming conventions are local, and the same operator can use several conventions.

Location information matters operationally because DNS and content-delivery systems use resolver or client location when choosing service replicas. Foundational evaluations showed both the utility and limitations of DNS-based server selection [18]; later work examined how public resolvers interact with global traffic management [19]. Cellular measurements further show that distance between a client and its resolver can change replica selection and resulting performance [20]. Current load-balancing documentation continues to expose geographic and latency-based steering as operational inputs [7]. These applications make infrastructure location useful, but they do not make an inferred city label authoritative.

Aleph extends PTR-based work by using language models to synthesize operator-specific extraction rules and decode non-standard hints [21]. Bet reorganizes this process around reusable operator-local contexts and separates semantic review from independent latency-based plausibility checks [22]. Together, Aleph and Bet provide a scalable method for producing inspectable city-level claims for covered, PTR-bearing service-facing infrastructure and rejecting claims inconsistent with independent evidence.

The proposed GeoCorrect study [23] extends this research line from operator-encoded infrastructure names to commercial end-user IP geolocation claims. Ordinary client traffic supplies passive RTT observations; the study sends no measurement probes and does not assume a comprehensive city-level reference dataset. Instead, it asks whether a provider claim is consistent with RTT constraints and when RTT alone, or a time-aligned customer correction together with RTT, supports a bounded alternative. This changes the evaluation target from universal database accuracy to the defensibility of a specific audit outcome for a stated operational use.

2.4 Provider identity, delegation, and dependency concentration

The autonomous system is the standard unit for interdomain routing measurement, but an ASN identifies a routing domain rather than a company, provider, or independent failure domain. One organization can operate many ASNs, while one ASN can carry several services, customers, or administrative units. Actor-level analysis therefore depends on a separate mapping step.

CAIDA's AS-to-organization data [24] and extensions such as AS2Org+ combine registration and PeeringDB evidence to identify candidate sibling ASNs [25]. Chen et al. improve sibling inference by integrating additional organizational signals [26], and Borges uses free-text PeeringDB fields and website evidence to recover relationships missed by structured records [27]. Each mapping has different coverage, freshness, and authority; agreement is useful evidence of relatedness, not proof of legal ownership or unified operational control.

Identity assumptions affect measurements that aggregate routing objects into actors. AS-relationship and customer-cone inference deliberately models routing relationships at the ASN layer [28]. When analysts subsequently interpret those results as provider concentration, supplier diversity, or independent failure domains, sibling ASNs can create false independence. Aggressive organizational merging can create the opposite error, false unity, by collapsing operationally distinct units.

This matters to DORA Article 29's concern with the same, closely connected, and difficult to substitute providers, and to organizations assessing correlated dependencies. The register,

however, represents providers through LEI/EUID and records the financial entity's service-specific substitutability assessment; an ASN grouping can challenge a network-diversity interpretation but cannot validate either declaration. Appendix A makes that comparison explicit.

Application-layer dependencies add role ambiguity. Studies of third-party dependency show substantial variation in Web-service centralization across countries [29], and work on government hosting shows that DNS and hosting choices expose different degrees of external-provider involvement [30]. Email makes the layering especially clear.

Large-scale MX studies classify mail-service provider usage from DNS and active observations [31], while Foster et al. show that provider-mediated email security depends on which infrastructure actually processes messages [32]. Yet an MX record, the network announcing its address, a Simple Mail Transfer Protocol (SMTP) greeting, and a Sender Policy Framework (SPF) authorization describe different service roles. None alone identifies mailbox storage, legal ownership, or every subcontractor.

Miscounting addresses the first gap by recomputing concentration, routing, RPKI, topology, and churn results under six identity models [33]. Its ongoing analysis asks whether an actor-level conclusion is stable under plausible ASN groupings. GOVT-MX addresses the second gap by composing MX configuration, application identity, routing and registration, geolocation, and SPF authorization according to their evidentiary authority [34]. This supports bounded attribution of inbound placement and outbound delegation while preserving the distinction between provider identity, observable control, and authorized service roles.

2.5 Request-path observability and cooperative verification

DNS resolution is often described as a client querying a recursive resolver, which then queries authoritative servers. Deployed resolution can instead include local stubs, forwarders, shared caches, public-recursive frontends, and resolver egress systems.

Measurements of client-side DNS infrastructure document this heterogeneity [11]; collaborative resolver measurements identify recursive systems that rely on other resolvers [35]. Transparent forwarders further separate the address contacted by a client from the system that performs recursion [36]. Interception can even preserve the apparent source address while substituting another resolver [12]. At ecosystem scale, concentration studies use authoritative-side observations to characterize which recursive providers generate visible DNS traffic [37]. These studies can reveal endpoint differences and population-level dependencies, but an endpoint pair does not enumerate the intermediaries between them.

Existing DNS protocol mechanisms address adjacent problems. Extension Mechanisms for DNS (EDNS(0)) defines an extensible option mechanism for DNS messages [38]. The Name Server Identifier (NSID) option lets a responding name server return a server-chosen identifier, but the identifier's semantics are intentionally operator-defined and the option does not provide a transitive, integrity-protected resolver path [39]. DNS over HTTPS protects DNS exchanges using HTTPS [40]; Oblivious DoH separates the client address visible to a proxy from the query visible to a target [41]. These protocols improve channel confidentiality, authentication, or privacy under their specified threat models. They do not ask unknown forwarding resolvers to identify themselves.

Distributed tracing provides a useful application-layer comparison. With OpenTelemetry, instrumented services propagate context that lets tracing systems assemble observations from cooperating processes into one trace [8]. That model is powerful when an organization controls or coordinates the participating services; it also makes the participation assumption explicit. An external DNS client generally lacks that shared instrumentation and may not know which organizations participate

in resolution.

DNSTRACING compares the client-facing resolver with the source observed at an authoritative server, treating disagreement as endpoint evidence rather than a complete path [42]. Cooperating resolvers can report participation through an EDNS option; response-bound attestations then let clients check received assertions, ordering, query and selected-response binding, and freshness under stated key and replay assumptions. The prototype authorizes ASN-to-key bindings with an RPKI object under a private test trust anchor; RPKI does not certify the country assertion. Disclosure adds information and attestation protects received reports. Neither establishes complete participation or detects arbitrary whole-option stripping.

Chapter 3

From partial evidence to bounded conclusions

The studies follow a common approach to answering the questions in Chapter 1. We first identify what a network observation could tell us about a location, operator, service role, or request participant. We then test that interpretation with evidence suited to the question and state what remains unresolved. For example, a hostname can suggest a place, an RTT can constrain whether that place is physically possible, and an operator record can explain the naming convention. Combining these sources helps only if we keep track of what each source establishes.

3.1 Observation, interpretation, check, and conclusion

To let another researcher inspect an inference, we record five elements: what we observed, when, and in which population; the property we want to determine and the unit we analyze; the rule or model used to interpret the observation; the check applied to that interpretation; and the conclusion with its limits. This record lets a reader distinguish new observations from a new interpretation of the same observations.

The check depends on the question. A location constraint may rule out one candidate while leaving several others possible. Comparing plausible ASN-to-operator mappings reveals whether a concentration estimate depends on how the analyst groups ASNs. Attributing an email role requires evidence about that role: hosting, inbound handling, and sending authorization are different claims. DNS also exposes a limit of interpretation. If ordinary observations leave no distinguishing record of an intermediary, a cooperating resolver must supply new information before a recipient can check its origin and integrity. Table 3.1 summarizes these questions and checks.

Table 3.1: The question, output, and evidentiary limit of each study.

Study	**Question and supported output**	**Check and boundary**
Aleph and Bet	What geography can names reveal? Reusable extractors and candidate locations.	Benchmark recovery, operator semantic review, and RTT compatibility test different properties.
GeoCorrect	Can a consequential location claim be rejected or corrected? An exclusion or supported alternative.	Passive RTTs and applicable customer evidence; uncertainty and conflicts remain explicit.
Miscounting	How concentrated is infrastructure under explicit identity assumptions? Model-dependent estimates and trends.	Dated mappings over routing inputs; no complete operator census or proof of independent failure.
GOVT-MX	Who performs each visible email role? Separate inbound, hosting, and authorized-sending attribution.	Role-specific evidence; authorization is not observed sending or mailbox location.
DNSTRACING	Who else handled a query? Endpoint findings and cooperative participation reports.	Response-bound verification authenticates received reports, not complete participation.

3.2 Evidence classes and trust assumptions

Four kinds of evidence recur throughout the dissertation. *Protocol observations* record route origins, MX records, SPF authorizations, and resolver endpoints. *Operator-maintained metadata*, such as PTR names and service identifiers, uses conventions that may differ across operators. *Derived mappings and measurements* include ASN-to-entity associations, geolocation records, and RTT bounds. *Cooperative reports* contain statements that participating infrastructure supplies explicitly. For each source, we identify the population and time it covers and the assumptions needed to use it.

Each source has authority over particular claims. RPKI authorizes an ASN to originate a prefix [43]; it does not identify a service owner. SPF authorizes sending for a domain [44]; it does not establish that sending occurred. An RTT bound can reject a physically incompatible location without identifying the true city. Agreement between organization mappings is evidence of relatedness, not proof of legal ownership. Combining sources preserves these distinctions.

3.3 Inference, validation, and verification

We distinguish three operations that test different properties. In *inference*, a researcher or system applies an explicit model to connect observations to a claim. The model may interpret a hostname, assign an ASN to a group, or attribute an inbound mail service. If the observations permit several answers, the method must report those alternatives or decline to choose among them.

In *validation*, we test the claimed property with relevant evidence. Operators reviewing Bet's outputs check hint meaning; RTT measurements check physical plausibility. These checks answer different questions. Miscounting compares models to test sensitivity. GOVT-MX uses evidence about each mail role to assess the corresponding attribution.

In *verification*, a recipient checks specified integrity properties of a received statement. Resolver-Path checks signatures, report ordering, association with the query and selected response, and freshness under its key and replay assumptions. A report that passes these checks can still omit a silent intermediary or contain an unverified location assertion. Disclosure supplies information; attestation protects the information received under the protocol's assumptions.

3.4 Using the conclusions

The resulting outputs can guide a location dispute, expose sensitivity in a concentration estimate, distinguish email responsibilities, or authenticate a resolver's report. Their usefulness depends on the operational question. A country exclusion may settle one access restriction while leaving another undecidable. A shared ASN group may motivate investigation without establishing a common failure domain.

Regulatory auditors could use these outputs if they can link a contracted service to technical identifiers at the relevant time and consult parties with access to contractual and internal records. Appendix A explains this possible DORA application.

Part I

Location

This part asks how infrastructure names yield candidate locations at Internet scale and when independent measurements can reject a location claim without identifying a replacement. Aleph and Bet recover and interpret hostname hints; GeoCorrect tests consequential location claims using passive RTT evidence, keeping reported meaning separate from physical compatibility.

Chapter 4

The Aleph: reusable interpretation of operator-local hostnames

The Aleph learns reusable rules and mappings for operator-assigned hostnames. This chapter presents the pipeline, its coverage and initial validation, and the recovery benchmark that motivates Bet.

A DNS PTR record maps an IP address to a hostname. Operators often include geographic codes to support routing, troubleshooting, and capacity planning. The code may be a city, airport identifier, abbreviation, facility, neighborhood, or private mnemonic. Conventions differ across business units and legacy domains, and names can remain after mergers or relocation. A hostname is therefore operator-maintained evidence whose meaning must be interpreted, rather than a standardized declaration of physical placement.

A global dictionary can recognize a familiar token, but its meaning may depend on the surrounding hostname and network. An operator-specific parser extracts the relevant field from many records, and a mapping assigns it a candidate location. Round-trip time (RTT) measurements can then test whether that location is physically compatible with the observed delay. The studies below evaluate these steps separately: recovering geographic hints, interpreting their meaning, and testing the resulting locations.

4.1 From contextual interpretation to reusable artifacts

An LLM can interpret a hostname as text in context: for example, the token `chcgil` in an AT&T hostname suggests Chicago, Illinois. The output is a geographic hypothesis, not an observation of the device's position. Asking the model to interpret every PTR separately would repeat expensive inference over many instances of the same convention. The Aleph instead groups sampled hostnames by naming pattern, learns an extractor for each group, and maps the extracted hints to locations [21]. A group is a *naming class*; its extractor is a regular expression that captures the field containing a possible geographic hint. The pipeline reuses these rules and mappings across matching hostnames, so it need not invoke the model for each address.

The Aleph's initial one-shot design asked an LLM to generate regular expressions and map every extracted string to a location in the same step. This design failed because it combined two tasks: identifying where a geographic field appears in a hostname and interpreting what that field means. For a network such as AT&T, an apparent rule such as "three letters followed by a two-letter state code" misses longer codes, codes containing digits, and codes that omit a state. A broadly written regular expression improves recall but also captures access identifiers or interface numbers. The combined task therefore hides whether an error came from parsing or interpretation.

The Aleph separates the work into three stages within each autonomous system (AS), keeping the operator context available throughout:

1. The LLM groups sampled PTR hostnames by structural naming pattern to separate conventions associated with legacy domains, access technologies, and service types. The implementation supplies between 337 and 642 records, depending on hostname length and the model's context limit.

2. The LLM generates a regular expression from at most five examples of each class. The expression's capture group selects the candidate geographic hint.

3. The pipeline applies each extractor to a different sample and removes duplicate hints. The LLM then maps each distinct hint to normalized city, state, and country fields.

Additional samples expose more hints. Once the mapper interprets a hint, the pipeline can reuse that decision for every matching PTR record.

This decomposition makes the output inspectable. For example, records under `sbcglobal.net`, `ameritech.net`, `swbell.net`, and `bellsouth.net` can belong to different structural classes even though their addresses originate from the same AS. A reviewer can inspect the class examples, test the generated regular expression against the corpus, and then review a table mapping captures such as `austtx`, `chcgil`, and `dlstx` to Austin, Chicago, and Dallas. The system uses the LLM to synthesize a reusable parser and semantic mapping; it does not place the LLM on the data path for each address.

The published implementation uses GPT-4 Turbo with temperature zero and makes the prompts, input records, and intermediate outputs available for inspection [21]. Temperature zero reduces sampling variation under a fixed model version, but it does not make the model or its future versions deterministic. Reproducibility therefore depends on retaining prompts, examples, model identifiers, generated rules, and mappings rather than merely recording the model name.

4.2 Feasibility and coverage

We evaluated Aleph on a February 2024 OpenINTEL PTR snapshot [45]. To cover the networks responsible for 90% of the snapshot's PTR records while increasing geographic and network-type diversity, we selected 2,646 ASes. The set included the 2,043 largest ASes by PTR count, access networks selected using APNIC population estimates [46], large transit networks, and prominent content providers. Collectively, the selected ASes accounted for 1.16 billion PTR records and an estimated 84.31% of the Internet user population represented by the selection method [21].

Table 4.1: The Aleph's February 2024 evaluation population and outputs. Counts describe the selected 2,646-AS population, not all Internet addresses.

Quantity	Reported value
PTR records in selected ASes	1.16 billion
PTR records with extracted geographic information	224.2 million (19%)
ASes with at least one extracted geographic hint	1,551 (58%)
Generated regular expressions	4,910
Distinct geo-hints	16,108
Distinct city-level locations	6,025
Countries represented	206
Generation time and model cost	approximately 2 days; $500

Aleph extracted geographic information at scale, producing reusable rules and mappings in approximately two days (Table 4.1). Its outputs covered nineteen percent of PTR records in the selected population, with at least one geographic extraction in 58% of selected ASes. The first rate measures record coverage; the second measures how many selected ASes yielded any output. Neither describes coverage of all IP addresses or the share of each operator's infrastructure that Aleph can

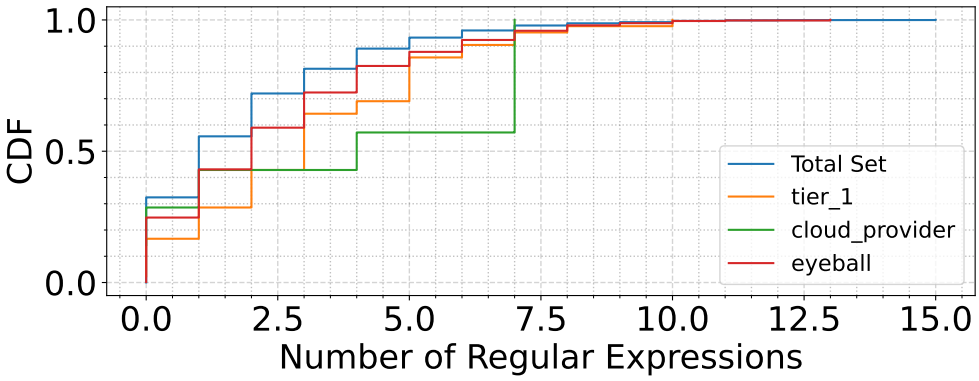


Figure 4.1: Generated regular expressions per provider, one per learned naming class, in The Aleph's evaluation. Curves show the complete evaluated set and selected network types; *eyeball* denotes access networks. Learned classes expose naming diversity rather than a ground-truth census of all operator conventions. Reproduced from [21, Figure 3].

locate. Networks without PTR records, opaque names, and networks outside the selected set remain beyond this evaluation.

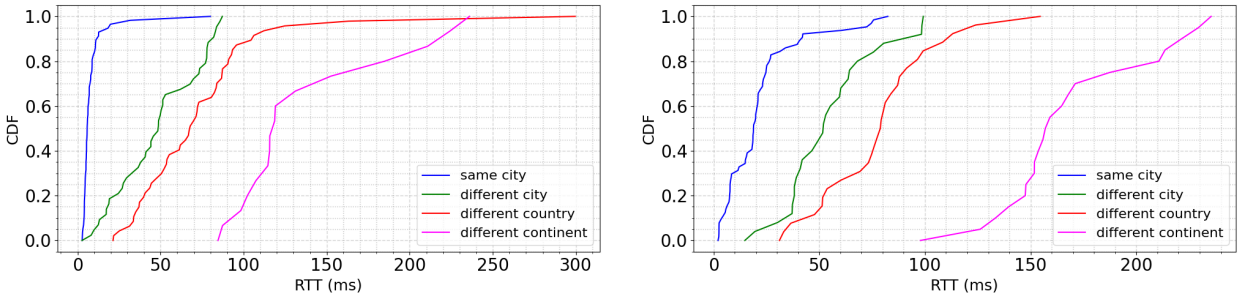
Most extracted hints used custom operator encodings, which a fixed geographic dictionary cannot fully describe. We classified nearly 66% of hints as custom encodings, compared with 16.68% International Air Transport Association (IATA) codes and 12.06% place names. Half of the measured networks used at least three naming classes, and one fifth used more than five (Figure 4.1). Large cities appeared under many distinct strings: we found 123 hints for Tokyo, 95 for Chicago, 74 for Paris, and 67 for Los Angeles. These counts show that the vocabulary to be interpreted is open-ended and depends on operator context.

4.3 Initial validation

We checked the inferred locations in two ways: operator records tested the meaning of hostname hints, and active RTT measurements tested physical plausibility. Comcast provided mappings for 24.5 million addresses in 3,265 prefixes that overlapped the snapshot. Among 210,926 addresses with city-level PTR inferences, 99.3% agreed with the provided city; state-only inferences agreed in the reported population. The errors included PTR labels that the operator data itself identified as stale or mislabeled. IIJ operators reviewed 31 inferred locations and confirmed 30; the failure mapped the ambiguous code `mtk` to Matsukawa rather than Mitaka. For Arelion, 38 of 47 extracted hints overlapped with a public looking-glass mapping, and all 38 overlapping mappings agreed [21].

These are strong checks for the reviewed operators, but they are not a random sample from the full 16,108-hint population and therefore do not estimate an Internet-wide accuracy rate.

The active experiment asked whether targets responded faster to probes near the inferred city. For selected access, transit, content, and cloud networks with responsive infrastructure, RIPE Atlas [47] probes measured RTT from vantage points in the inferred city, other cities, other countries, and other continents. We used the RTT distribution from the inferred city as the reference and compared it with each more distant group. Across the selected operators, Kolmogorov–Smirnov distances, which measure differences between distributions, generally increased with the distance from the inferred location. The reported comparisons were statistically significant at $p < 0.05$. This result



(a) Verizon

(b) AT&T

Figure 4.2: RTT distributions for selected responsive targets in two networks. *Same city* denotes RIPE Atlas probes in the city inferred from the target's hostname; the remaining categories place probes farther away. Lower RTTs from the inferred city support physical plausibility, not independent verification of city identity. Original panels reproduced from [21, Figure 7(a–b)].

supports aggregate physical plausibility. It neither proves an individual city label nor covers devices that do not answer probes. Figure 4.2 shows the published Verizon and AT&T examples.

Aleph also mapped more hints than Hoiho [16] in all seven networks of a controlled comparison [21]. We applied both systems to the same February 2024 population derived from CAIDA's Internet Topology Data Kit (ITDK) [48]. Aleph mapped 272 hints versus 29 for AT&T, 51 versus 10 for Claro Brazil, and 77 versus zero for China Telecom. The additional mappings included nonstandard abbreviations such as `fz` for Fuzhou and regional codes whose dictionary interpretation was an airport in another country. This comparison demonstrates the coverage available from contextual semantic inference, while the limited seven-network population prevents a general claim that every additional mapping is correct.

These evaluations establish that contextual interpretation yields reusable artifacts and useful coverage. They leave a different question unanswered: *how much of the geographic information available in a hostname corpus does Aleph recover?* A record with no output may contain no geographic hint, or it may contain information that the pipeline missed. Answering this question requires annotations independent of the learned artifacts.

4.4 Measuring the geographic information Aleph recovers

To measure what Aleph missed, we built a diverse 30-AS benchmark in which reviewers identified the geographic information present in selected names [22]. The May 9, 2025 sampling frame contained 18,502 eligible ASes. We divided them into three CAIDA ASRank [49] strata and, within each stratum, three tiers based on the number of registrable domains. We refer to these domains as second-level domains (SLDs). We selected three or four ASes per cell, with constraints on registration geography, and uniformly sampled 100 distinct canonical hostnames from each selected AS. This design supplies a manageable human reference for a heterogeneous billion-record corpus.

We excluded all 3,000 sampled hostnames from class learning, regular-expression (regex) induction, and hint-mapping inputs before running the pipelines. Four reviewers annotated geographic spans, groups of components that jointly form a hint, supported interpretations and their granularity, and structural naming conventions. They divided the work under a shared rubric and resolved disagreements by group vote; the manuscript does not describe a uniformly double-annotated or blinded sample. After excluding 37 unresolved records, the benchmark contained 2,963 scorable

PTRs: 506 with geographic references containing 509 hint groups, and 2,457 without geographic references.

We tested Aleph in three runs with a 2,500-hostname per-AS learning budget, keeping the model, inputs, settings, and post-extraction checks fixed. The scorer counts complete annotated groups and unmatched captured spans to assess extraction. It credits *finest-granularity recovery* only when the output also maps the hint at the reference's finest geographic level. This additional requirement tests whether the method recovers the available geographic detail; it does not verify the identity of the resolved place.

Aleph recovered geographic information at the finest annotated level for a mean of **70.6%** of the 506 geographic PTR references (mean 70.55%; sample standard deviation 7.66 percentage points across the three runs). The remaining shortfall motivates the Bet design in Chapter 5. The diagnostics identify concrete failure modes, but do not isolate how much each contributed to the shortfall.

4.4.1 Parsing and mapping can disagree on the same hint

In one recorded Aleph run for AS3136, the hostname `dwdmadwse150n.jobcenterofwi.com` yielded captures `mad` and `wi`, concatenated into the extraction key `madwi`. The mapper reread the hostname and emitted separate `mad` and `wi` entries. The composite therefore remained unmapped even though its components were recognizable. This is a mismatch between the parser's vocabulary and the mapping keys, not simply a failure to recognize a city [22].

4.4.2 Naming conventions vary within domains

A registrable domain provides useful context, but names within it can still follow several conventions. AT&T's `att.net` and `pacbell.net` use different layouts, while Orange's `abo.wanadoo.fr` suffix includes both abbreviated location codes and prefixed place names. Among 14.43 million IPv4 PTR records matching three examined structures under the latter suffix, 38.6% follow non-dominant structures. Using the domain alone as the naming class would therefore combine different layouts.

The held-out names span 243 AS–SLD groups. Among the 83 with more than two matches eligible for adjudication, 28 (33.7%) contain multiple human-labeled naming conventions. In run 1, reviewers assessed learned classes with at least two reviewed holdout members for incompatible geographic-field layouts. Ten of Aleph's 29 assessable classes mixed conventions; 254 of its 283 classes had insufficient reviewed members. These diagnostics show that learning classes across an entire AS can combine incompatible layouts. Because most classes lacked enough reviewed members, the counts describe the assessable subset rather than the frequency of mixed classes across every network or naming convention.

Chapter 5

Bet: structure-aware geolocation at Internet scale

Bet addresses two problems exposed by the benchmark in Section 4.4: a single AS can contain several naming conventions, and the mapper can interpret a different string from the one the extractor captured. Bet retains local domain context when learning and applying extractors, then asks the mapper to interpret the exact vocabulary those extractors produce [22].

5.1 Changing the inference unit

Aleph already supplies reusable class-specific extractors; Bet changes the context in which it learns and applies them and the inputs it gives the mapper.

For an AS a, let H_a be its PTR hostnames. Bet partitions H_a by SLD and learns multiple structural classes within that context. This preserves domain-specific patterns without assuming that a domain has only one layout.

Bet samples up to 2,500 distinct learning hostnames per AS after excluding evaluation holdouts. To represent domains in proportion to their size, it treats SLDs with at least 50 eligible hostnames as separate sampling strata and places smaller domains in a shared residual stratum. It allocates slots proportionally, using fractional remainders to determine the final allocation. For each SLD with at least 50 sampled hostnames, Bet asks the model to classify that domain's names separately; it combines the remaining samples in one classification request. The model then generates a regular expression from up to five examples of each class. Sensitivity experiments on a ten-AS subset of the annotated benchmark support these defaults.

The scanner applies the learned expressions within their SLD context. It normalizes the captured components, removes empty and numeric-only components, and concatenates the rest into a candidate hint. Before mapping, separate eligibility checks filter out shapes such as long strings containing digits or strings resembling hexadecimal values. Bet retains their extraction records, so filtering a candidate from mapping does not erase its extraction.

Within a run, Bet keeps one mapping key for each combination of method, ASN, and normalized hint. It combines the associated class identifiers and example hostnames under that key. Thus extraction uses SLD context, while mapping uses AS context: identical strings in different ASes can receive different interpretations, and millions of records within an AS can reuse one decision.

5.2 Evidence-bearing hint cards

Bet enumerates the full vocabulary produced by its extractors across the corpus before asking the model to map it. For each distinct hint, it creates a *hint card*: the exact extracted string, up to two example hostnames (called witnesses), and optional geographic constraints derived from RTTs. Bet samples witnesses uniformly without replacement from distinct hostnames associated with the method/ASN/hint key. The mapper returns normalized city, administrative-region, and country fields for that key [22].

The card keeps extraction and mapping aligned. For the `madwi` case in Chapter 4, it would ask the model to interpret the composite key, rather than substitute separate entries for its components.

Bet retrieves the card's evidence from outputs already produced by the pipeline. Its default *lexical mapper* uses the hint and hostname context without requiring an external geographic dictionary at inference time.

Optional RTT measurements help the mapper distinguish candidate locations that share a code. For a target IP i and vantage point v, Bet converts RTT to a distance bound and intersects the bounds from the set of vantage points V:

$$r_{v,i} = \frac{2c}{3}\frac{\mathrm{RTT}_{v,i}}{2}, \qquad P_i = \bigcap_{v \in V} B(v, r_{v,i}), \tag{5.1}$$

Here c is the speed of light, $B(v, r)$ is the geographic disc centered on the vantage point, and P_i contains locations compatible with all applicable bounds. The factor $2/3$ approximates propagation speed in fiber, and dividing RTT by two estimates one-way delay. Bet then intersects the regions for IPs carrying the same hint to obtain a shared footprint. If those regions have an empty intersection, it uses the member region with the smallest area and records that fallback. This occurred for 0.066% of IPs in the May 2025 run. Hints without usable measurements rely on lexical interpretation alone. The geometry constrains physical plausibility, while the hostname supplies the candidate geographic meaning.

5.3 Implementation and reproducibility

Bet ingests weekly PTR snapshots for the non-bogon IPv4 address space and a curated, de-aliased IPv6 hit list [50]. The collection process resolves IPv4 and IPv6 addresses separately; the longitudinal analysis then merges the results and removes duplicates. We evaluated the model-based stages with Gemini 2.5 Flash using greedy decoding, a fixed seed, and a fixed order of examples. Bet batches hint cards into windows of at most 100,000 tokens. These controls limit variability but do not make a hosted model deterministic across requests or versions [22, 51].

All reported RTT experiments used the commercial provider's 1,131 probes across 143 countries and 308 ASes. The measurement process checks reachability weekly from four randomly selected vantage points. For targets that answer pings and have prior observations, the process selects the five lowest-RTT and five highest-RTT vantage points, then refines the feasible region using three probes selected within it. Random probes fill gaps when prior observations or eligible vantage points are unavailable. Echo replies and TTL-exceeded hop responses supply RTTs, which the process associates with the address that responded. RIPE Atlas [47] could supply replacement measurements for Bet, but did not supply data for the reported Bet evaluation.

The manuscript documents retained prompts, model settings, schemas, regexes, hint cards, sampling rules, and run outputs. The commercial RTT feed cannot be redistributed. Public PTR inputs and lexical artifacts support replication of the naming pipeline, while reproducing the physical-plausibility and RTT-informed experiments requires a new measurement feed. We still need to complete the provenance records for the paired-recall bootstrap and validation archive: per-AS counts, scripts, run and scorer versions, and learning-input identifiers. This work will make the reported experiments easier to inspect and reproduce; the benchmark itself is complete.

5.4 Returning to the same recovery benchmark

Bet increased mean finest-granularity recovery recall to **90.9%** (90.91 ± 4.99), compared with Aleph's 70.55 ± 7.66, on the same 506 geographic references within the 2,963 scorable PTRs. Both pipelines

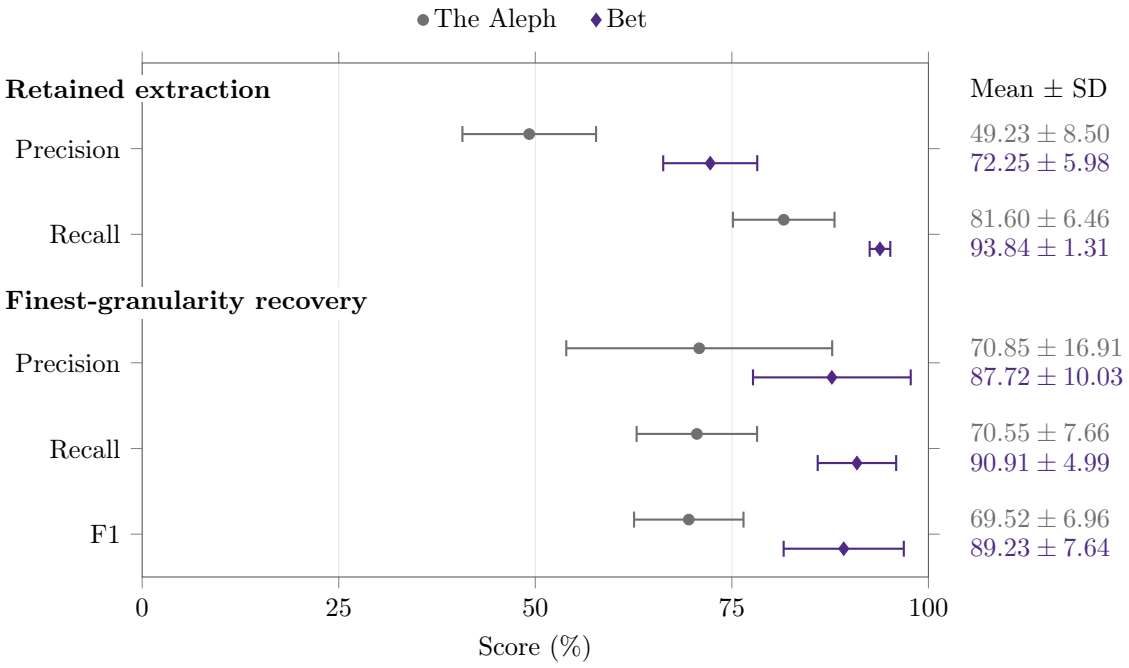


Figure 5.1: Completed Aleph–Bet comparison on 2,963 scorable PTRs from 30 ASes. Points and whiskers show mean ± sample standard deviation of three run-level scores (%). Extraction uses 509 annotated hint groups; recovery uses 506 geographic PTR references. Scoring pools counts within each run before averaging scores across runs. Recovery tests annotated hints and granularity, not independently verified place identity. Data from [22, Table 3(a)].

used the same model, per-AS learning budget, frozen holdouts, and post-extraction checks. Figure 5.1 reports precision, recall, and F1 with their run variability [22].

Mean recovery F1 increased by 19.70 percentage points using the unrounded run-level scores. Only 13 of the 30 ASes supplied geographic references, and three ASes accounted for most of the recall gain. Across the three runs, counting a hostname separately in each run, Bet had 138 missed recoveries versus 447 for Aleph and 185 false positives on names without annotated geographic references versus 403. The comparison evaluates the complete pipelines: classification, regex induction, matching scope, capture processing, and mapping. It supports improved recovery in this benchmark without isolating SLD partitioning or hint cards as the sole cause.

Bet also produced mapped output for more records in a separate availability comparison across the same cohort. After excluding all 3,000 holdouts, this comparison covered 3,101,348 PTR rows and unique IPs per run. Bet produced mapped output for $84.6 \pm 5.8\%$ of these records/IPs versus $78.9 \pm 5.6\%$ for Aleph, and at least one output for $63.3 \pm 0.0\%$ versus $56.7 \pm 3.3\%$ of the 30 ASes. These rates measure whether an output exists, without estimating its correctness across the cohort.

Bet combined incompatible naming layouts less often in the subset reviewers could assess. In run 1, it had six mixed classes among 35 assessable classes (17.1%), compared with Aleph's 10 of 29 (34.5%). Most learned classes lacked sufficient reviewed holdout members: 259 of 294 for Bet and 254 of 283 for Aleph. The comparison therefore characterizes the assessable subset, rather than estimating coherence across every learned class.

5.5 Corpus results, validation, and temporal reuse

The benchmark establishes how much annotated information Bet recovers. The remaining evaluations ask how far its extractors reach, whether it interprets hints correctly, whether inferred locations satisfy RTT constraints, how they compare with database labels, and whether the extractors remain useful over time. Each check answers a different question (Table 5.1); the populations and results below follow the updated manuscript [22].

Table 5.1: Evidence channels used to evaluate Bet.

Evidence	Supports	Does not establish
Annotated 30-AS benchmark	Recovery of annotated hints and their finest available geographic level	Resolved place identity or an Internet-wide accuracy ranking
Operator review and public artifacts	Correct interpretation of reviewed operator-local hints	Accuracy of every IP sharing a hint
RTT feasibility regions	Compatibility with measured latency bounds in measured subsets	Exact position or coverage of unmeasured IPs
Commercial database comparison	Convergence and disagreements requiring adjudication	Ground truth for infrastructure location
Corpus extraction analysis	Coverage from complementary learned patterns	An isolated causal effect of one pipeline component
Longitudinal snapshots	Reuse of learned extractors and hint vocabulary	Continued correctness of stale location mappings

5.5.1 Internet-scale execution

Bet processed 1.58 billion PTR records in the May 2025 extraction evaluation, covering 14,616 IPv4 ASes and 4,738 IPv6 ASes and approximately 99.9% of the corpus. An AS can appear in both populations; these analyses therefore treat each AS/address-family pair as a network. The run produced 107,248 usable class instances with executable extractors for IPv4 and 16,870 for IPv6. These included 102,039 and 16,483 distinct executable extractors, respectively. Applying their union yielded retained hints from 728.6 million IPv4 records and 127.4 million IPv6 records [22].

Extraction coverage was 62.13% of non-null IPv4 PTRs and 61.23% of non-null IPv6 PTRs, or approximately 54.2% of all raw records across both families. Equal-weighted per-network median coverage was lower: 50% for IPv4 and 28% for IPv6. These differences distinguish aggregate record coverage from the experience of a typical observed network. Coverage did not increase consistently with observed PTR footprint, and 12.0% of eligible IPv4 networks and 35.3% of eligible IPv6 networks yielded no hints.

A small share of distinct candidates accounted for most records with mapped output. Bet had mappings for 7.80% of distinct IPv4 AS–candidate pairs and 20.37% of IPv6 pairs. Because many records reuse those candidates, these mappings covered 82.63% and 98.98% of records yielding retained candidates, respectively. The pair counts measure how much vocabulary has a mapping; the record counts measure how widely those mappings apply. Neither measures whether the mappings are correct.

The corpus run completed in under 18 hours at approximately 1,050 ASes per hour and an inference cost of approximately $5,000. Per-network cost still varies with SLD diversity, naming-class count, and candidate-hint vocabulary. The pipeline limits model work by using a conventional

scanner for repeated records and invoking the LLM for learned classes and distinct hints.

The May 2025 run recovered approximately 1.35 million globally distinct hint strings, with custom encodings comprising 77.63% of outputs. A separate mapping-evaluation population contains 507 million PTRs with geographic hints, including 222 million city-level PTRs; its corresponding IP counts are 531 million and 228 million. The city-labeled IPs span approximately 52,700 cities in 229 countries and territories. The vocabulary includes multilingual place names, facilities, neighborhoods, and administrative regions. A specific-looking name can still denote an operational region rather than the addressed device's exact placement.

5.5.2 Complementary coverage

Using all learned extractors covered 80.12 million more records across IPv4 and IPv6 than retaining only the best existing extractor for each network. For this comparison, we selected each AS/address-family instance's highest-yield executable extractor after observing its coverage and compared it with the union of that instance's extractors. Even if the single extractor mapped every record it captured perfectly, any choice of one existing extractor per instance would leave at least 63.54 million currently mapped records uncovered. The result shows that learned patterns provide complementary coverage; it does not test an optimally redesigned single extractor.

5.5.3 Semantic correctness

Operator feedback and public records supported 662 of 672 scored hint mappings, or 98.5% (Wilson 95% interval: 97.3–99.2%). The complete reviewed set contained 689 hints: 662 correct, 10 incorrect, and 17 unknown. We excluded unknown verdicts from correctness scoring. Seven operators reviewed their distinct hints, inferred locations, and representative PTRs directly. For two providers, we checked cloud-region mappings against public documentation [52, 53]; for Arelion, we checked the subset of hints covered by its looking glass [54].

The scored mappings label 70.80 million IPv4 addresses: 24.94 million for direct-feedback sources and 45.86 million for public-artifact sources. Sixty-five percent of reviewed hints were custom codes. The address footprint shows reuse of interpreted symbols, not independent validation of every IP. Participation and available documentation determine the reviewed population; the score therefore applies to these mappings rather than the full vocabulary. The errors include `san` interpreted as San Antonio instead of San Diego.

5.5.4 RTT compatibility

Most checked city labels fell within their IP addresses' RTT-feasible regions. To test this physical compatibility independently of lexical interpretation, we fixed Bet's default lexical labels before comparing them with IP-level RTT polygons. Of the 228 million city-labeled IPs in the mapping-evaluation population, a polygon-area threshold of 50,000 square kilometers retained 51.6 million. Their inferred points lay within the polygons in 93.7% of IPv4 cases and 87.1% of IPv6 cases.

Tightening the area threshold to 2,500 square kilometers retained 2.3 million IPs across approximately 1,900 ASes, with 91.1% IPv4 and 84.0% IPv6 consistency. Smaller polygons impose tighter geographic constraints but leave fewer addresses available for evaluation. These rates measure compatibility with latency constraints, rather than city-level accuracy.

A separate sensitivity experiment tested whether RTT context could help the mapper resolve ambiguous names. We supplied geometry during mapping only when polygon area was below 50,000 square kilometers. Twelve percent of extracted hint strings had multiple plausible interpretations; usable geometry supported a more physically plausible choice for 74% of these cases. In one

example, RTT context selected Chicago rather than the lexical-only Christchurch interpretation of `br02.chc01.as3491.net`. Consistency stayed equal or improved for 95% of the 14.6 thousand IPv4 ASes. Because geometry contributes to both selection and scoring, this is a sensitivity result rather than independent validation of the RTT-informed locations.

5.5.5 Commercial-record disagreement

Most Bet labels were close to commercial database labels, but a substantial tail of disagreements remained. We compared approximately 227 million overlapping IPs with city-level labels against a May 9, 2025 commercial database. The median distance between labels was 11.76 km; 66.6% were within 50 km and 83.5% within 500 km, while 9.4% differed by more than 1,000 km. Some large disagreements paired an operator-wide administrative location with a regional PTR code, such as Seattle versus Mumbai for AWS's `bom50`. These cases motivate adjudication rather than automatic corrections. Fewer than 0.01% of the greater-than-1,000-km disagreement population responded to the available ICMP probes, limiting RTT verification of that tail.

To adjudicate a disagreement, we would need time-aligned evidence linking the IP to its serving site, such as applicable operator records, or delay constraints that rule out a candidate. The lexical interpretation and the commercial label alone leave physical placement unresolved.

5.5.6 PTR changes and reusable extractors

LLM inference is costly, but a changed PTR string need not require a new extractor. In a schematic rename from `r1.qzfj.example.net` to `r9.qzfj.example.net`, the same rule can still capture the second label, `qzfj`. The record changes while the extraction convention and hint remain usable. A different rename can remove the location field altogether. The temporal question is therefore which changes require new rules, which only add vocabulary, and which leave the artifacts reusable.

We studied 42,355 ASes present in snapshots from the second week of May in each year from 2022 through 2025, totaling 1.6 billion distinct hostnames. For each pair of consecutive snapshots, we classified IPs as retaining the same PTR (survival), changing PTR (churn), disappearing (death), or newly appearing (birth). The first three categories partition the earlier snapshot's IPs, and we divided each count by that snapshot's size. We used the same denominator for birth, so its rate can exceed one.

Across all transitions, 48.4% of networks maintained survival above 0.90 and churn below 0.10, while 51.6% experienced at least one episode with death plus churn above 0.20 or birth above 0.30. These criteria can overlap: a network can retain its existing records while adding many new ones.

Bet uses three signals to decide whether a new snapshot requires new extractors or only additional mappings. For each AS, it applies the most recently trained extractors to the next snapshot. A Z-test at $\alpha = .05$ compares parsed-record coverage with coverage at the last training. Regex survival S measures the fraction of stored extractors that still match at least one current record. The third signal identifies hints absent from the cumulative vocabulary seen so far.

Bet regenerates extractors only when $S < 0.9$ *and* coverage decays. When $S \geq 0.9$, coverage is stable, and unseen hints exist, it keeps the extractors and maps the new hints. Otherwise it reuses the artifacts, including when the diagnostics give mixed signals. This policy checks whether the rules still apply and whether the vocabulary has grown; it does not test the continuing geographic correctness of stored mappings.

Most ASes in the reuse experiment retained their initial extractors across all three follow-ups. We selected the 100 ASes with the largest arithmetic mean PTR counts over the four years, representing 1.05 billion distinct hostnames or approximately 65.6% of the longitudinal population. Starting with

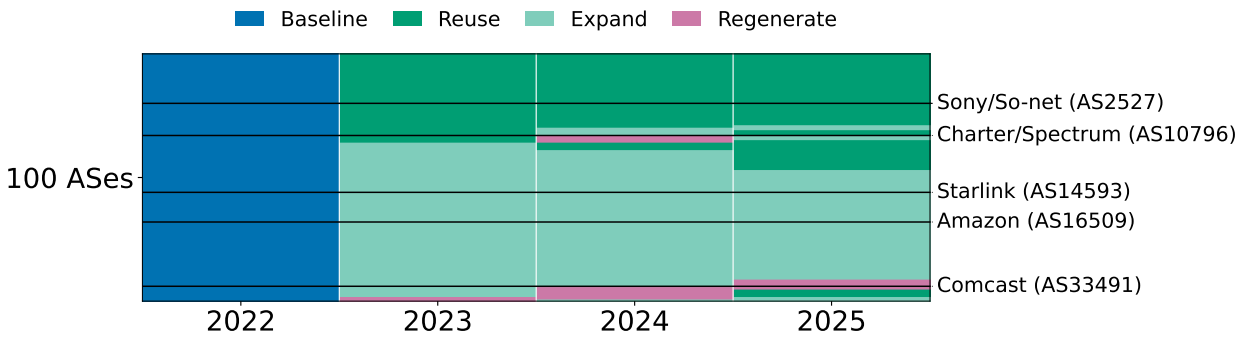


Figure 5.2: Update decisions after learning the initial extractors in 2022. Each row represents one of the 100 ASes selected by largest mean PTR count over 2022–2025. Seventy-eight retained their initial extractors through all three follow-ups, allowing vocabulary expansion. Reuse tests extractor applicability, not continued geographic correctness of stored mappings. Reproduced from [22, Figure 9].

2022 extractors, 78 ASes avoided full regeneration through all three 2023–2025 follow-ups, allowing vocabulary expansion. Annual counts averaged approximately 93 ASes avoiding regeneration and seven requiring it (Figure 5.2). Using per-AS May 2025 costs, the paper projects approximately $800 per year for selective updates across all 42,355 ASes versus $10,000 for full regeneration. We did not execute this full-population cost comparison directly.

The observed cases illustrate reuse, vocabulary expansion, and relearning. Sony/So-net (AS2527) retained all eight extractors with unchanged vocabulary. Starlink (AS14593) retained its extractors while expanding from eight to sixteen hints, consistent with its growing public PeeringDB footprint [55]. Charter/Spectrum (AS10796) retained roughly 8.6 million PTR records, but 8.29 million IPs changed from Road Runner names to `syn-...res.spectrum.com`. Those names removed the location-bearing tokens and triggered relearning. Regeneration cannot recover geography absent from the new names; the record changes do not identify every underlying operational cause [22].

5.5.7 Scope of the hostname results

Bet's completed benchmark supports recovery of annotated geographic information within the selected cohort, while corpus-scale extraction demonstrates how far its learned patterns reach. Addresses without observed PTRs remain outside this method's visibility. Among observed names yielding no hints, the current study does not determine how often geographic information is absent versus missed by the extractor. A separate original mapping evaluation assigned city-level labels to 228 million of 531 million IPs with extracted hints (42.9%); this denominator is neither all Internet addresses nor the broader extraction-characterization population.

Correct interpretation of a name also leaves physical-location uncertainty. PTRs may be stale, misleading, or refer to an interconnection or aggregation site. Operator evidence and latency constraints can help adjudicate particular cases; neither the large mapping footprint nor high benchmark recall removes that distinction. Hosted-model changes can alter learned artifacts even with fixed prompts, making retained inputs and outputs important for review and re-execution.

The completed studies evaluate recovery, class coherence, coverage, semantic correctness, and temporal reuse through reusable naming structures and mappings. Artifact-provenance records remain unfinished. Broader active measurement and region-guided probes remain optional extensions.

Chapter 6

GeoCorrect: passive evidence for disputed location claims

Aleph and Bet point toward candidate locations. A service using a location label must answer a further question: is that label supported well enough for the decision it makes? Return to the payment-API example. A country label can wrongly block an endpoint that meets a geographic rule, or admit one that does not. Labels can become stale after reassignment or relocation, and different sources may describe headquarters, regional infrastructure, or the endpoint itself. Customer disputes make these consequences visible, but a disputed label does not by itself reveal the correct physical location.

GeoCorrect asks when passive measurements can reject a provider's location label and when the evidence also supports a replacement [23]. Its input comes from ordinary client traffic observed at Cloudflare, without active measurement probes. The design uses a continuous, uniformly random one-percent sample of HTTP requests across a reported operational footprint of 765 points of presence in 337 cities. Sampling requests means that an address's chance of appearing depends on its traffic; this is not a uniform sample of unique addresses.

For each observed address and receiving site, the measurement input contains the minimum RTT in each hour. The site's coordinates locate the measurement origin, not the client. The pipeline analyzes an *address-window*: one address during a specified observation interval. It aligns the applicable RTT minima with the provider's dated location claim and, when available, a customer report. It then assigns an outcome that distinguishes a plausible claim, an exclusion, a supported replacement, or unresolved evidence.

6.1 Excluding a claim without locating the endpoint

GeoCorrect converts each RTT to a maximum-distance region around the known receiving site, under the declared propagation and measurement assumptions. A short RTT can rule out distant locations even when many nearby locations remain possible [56]. The pipeline intersects the applicable regions for an address-window to obtain P_i, and compares that intersection with the full claimed geographic region G.

The pipeline excludes G only if $P_i \neq \emptyset$ and $P_i \cap G = \emptyset$: some locations remain physically possible, but none lies in the claimed region. Testing only a representative city coordinate would not suffice to reject its entire country. If P_i is empty, the measurements or assumptions conflict; the pipeline must preserve that conflict rather than infer that the endpoint is nowhere.

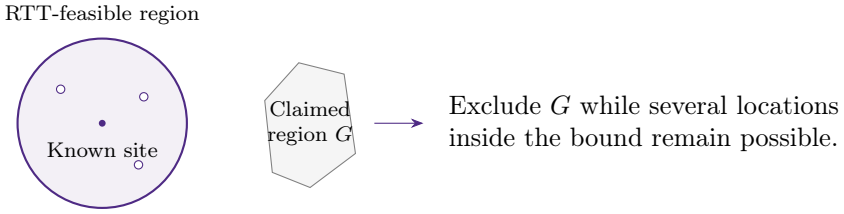


Figure 6.1: Schematic exclusion without replacement. The entire claimed region lies outside a nonempty feasible region. Open circles illustrate possible endpoint positions, not observed locations.

This asymmetry makes a narrower operational answer possible (Figure 6.1). If an illustrative API blocks only country B, excluding all of B can settle that geographic restriction without identifying the client's actual country. Other access conditions may still apply. Conversely, an allowlist requires support for membership in its allowed set; excluding one wrong country is not generally enough. A historical access decision also requires its policy version and request-time evidence, which hourly RTT minima alone do not supply.

A replacement requires stronger evidence than an exclusion. The proposed rules allow either an alternative tightly constrained by RTTs alone or one supported jointly by a time-aligned customer correction and RTT constraints. The customer report supplies a candidate to test, not physical ground truth. The pipeline's six final outcomes distinguish plausible provider claims, RTT rejection without replacement, the two correction paths, conflicting evidence, and insufficient evidence. Chapters 11 and 12 specify the decision rules and completion criteria.

6.2 Implemented pilot and remaining empirical questions

We exercised the passive aggregation, geometry, and classification pipeline in an hourly pilot [23]. The pilot establishes that these components work together; the proposed full-day and month-long evaluations will measure their outcomes at broader scope. The pilot does not establish a full-population exclusion rate, correction rate, or location-accuracy claim. Combining observations within a window assumes that they concern a stable physical endpoint. Reassignment, mobility, anycast, or shared egress can violate that assumption, so the analysis must report the resulting conflicts rather than force a location.

The proposed evaluation will use three populations to answer different questions. Operational address-windows will show how often the pipeline can exclude or replace provider claims and how often evidence is unavailable. Independently located stable controls will test erroneous exclusion within their covered population. Customer disputes and linked policies will identify the reported requirement and whether exclusion would settle it. Control error rates and complaint frequencies do not automatically generalize to all observed addresses. Over time, the analysis will report persistent outcomes, transitions, and coverage together; an address disappearing from the sample does not establish that its endpoint moved.

We propose a recurring feed to IPinfo that will report time-stamped exclusions and supported replacements as distinct outcomes. The workflow will compare new provider records with fresh passive evidence to identify recurring and consequential disagreements. It will not treat previous inferred corrections as independent validation. This feed remains a proposal: we have not deployed it or established that IPinfo has adopted the proposed corrections.

The location studies show how scalable interpretation and independent evidence can narrow a claim without resolving exact placement. GeoCorrect's remaining work will measure when exclusion is useful, how coverage limits it, and how often it errs on independent controls. The next part asks which operator or service actor an observed identifier represents.

Part II

Operator identity

Network identifiers can split one operator across several apparent actors or conceal several service actors behind one identity. Miscounting tests how operator groupings affect concentration and change, while GOVT-MX separates inbound handling, hosting, and sending authorization in government email.

Chapter 7

Miscounting: grouping network identifiers into operators

Counting network identifiers does not necessarily count the organizations operating a service. Two errors explain why. *False independence* occurs when several ASNs represent one operator: counting each ASN separately splits that operator's resources and dependencies across several apparent actors. *False unity* occurs when one network identity conceals several service actors: the network hosting an endpoint need not administer the application or every tenant behind it. Correcting one error does not automatically correct the other.

Miscounting the Internet asks how estimates of concentration and change over time depend on which ASNs an analyst groups as one operator [33]. Preliminary rankings and origin-change results show that the grouping matters; we still need to audit the mappings and validate the longitudinal concentration analysis. GOVT-MX addresses the complementary problem at the service layer. It asks which actors handle inbound government email, host that infrastructure, and have authorization to send on a domain's behalf [34]. Together, the studies make the choice of actor an explicit part of the measurement.

7.1 From routing identifiers to an identity model

An ASN identifies an autonomous routing system. It is the appropriate unit for observed route origins, AS paths, and Route Origin Authorizations (ROAs) [43]. Those protocol statements do not identify an independent provider, corporate owner, customer, or application operator. An analyst who treats every ASN as an operator therefore adds an identity assumption that the routing evidence alone does not support.

An *identity model* records that assumption as a dated mapping from ASNs to proposed operator groups. The mapping combines registry and WHOIS-derived organization records, operator reports, and grouping rules to associate ASNs with a group. Figure 7.1 illustrates this step. CAIDA AS2Org [24] and Borges [27] provide two such inferred organization mappings. They can produce different groups because they use different source records, normalization procedures, names, and update schedules. Neither directly observes complete operational control.

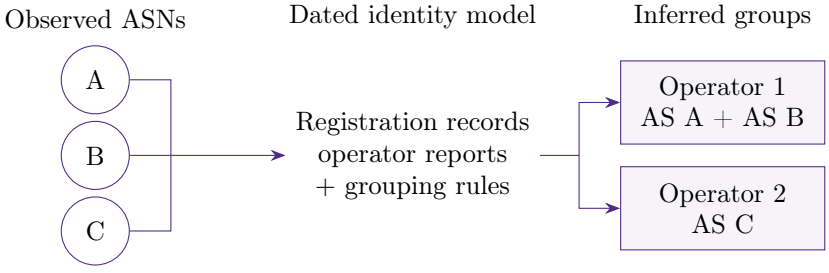


Figure 7.1: An identity model associates ASNs with proposed operator groups. The schematic keeps the routing identifiers fixed while changing their actor-level interpretation.

The relevant grouping depends on the question. A regional Internet registry (RIR) may record one resource holder while another organization announces the prefix. A parent company can own

subsidiaries with separate network operations, while distinct legal entities can coordinate one network. Application providers can run within a cloud, CDN, mail, DNS, or security provider's address space. Registry country, operator country, infrastructure location, and jurisdiction can differ. An acquisition record supports common ownership; a shared support page may support coordinated administration; an MX endpoint's origin ASN describes network placement. These are not interchangeable claims.

Miscounting uses *operator* to mean an administrative or operational grouping that runs multiple ASNs together closely enough that treating them as independent would misstate the measurement. This unit is narrower than ultimate ownership and broader than one ASN. Grouping the ASNs does not assign every customer or service within them to that operator.

Regional networks, retained acquisition identities, and separate mobile, residential, backbone, or cloud products can create false independence. For a fixed resource population, merging verified sibling ASNs—ASNs associated with the same operator—can only increase or preserve the share held by the largest k entities (top-k share). It can only decrease or preserve effective diversity, a measure of how widely those resources are distributed:

$$\text{top-}k\ \text{share}_{\text{operator}} \geq \text{top-}k\ \text{share}_{\text{ASN}}, \qquad N_{\text{eff,operator}} \leq N_{\text{eff,ASN}}.$$

The grouping can also reveal that an apparent external relationship or origin change connects ASNs within one mapped operator. These directional claims require justified merges and the same total resource population before and after grouping. False unity requires a different analysis: separating tenants or service roles within an ASN can change attribution and concentration in either direction.

7.2 Preliminary sensitivity motivates a longitudinal question

The choice of identity mapping changes the reported concentration of routed IPv4 addresses. In its final-snapshot comparison, the imported Miscounting manuscript reports that the ten largest groups account for **26.4% under CAIDA AS2Org** and **30.5% under Borges** [33]. Both analyses start from the same routing input. The resource measured here is routed IPv4 address mass: the addresses represented by the routed prefixes, subject to the study's weighting policy. Figure 7.2 displays the two shares. The difference motivates the central question: *which concentration trends survive plausible interpretations of network identity?*

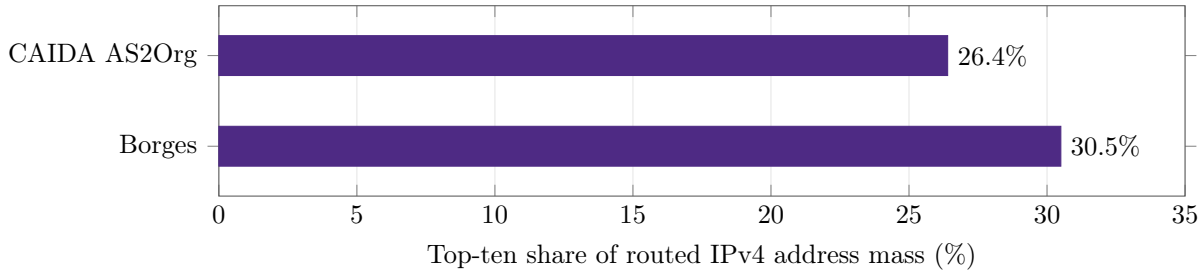


Figure 7.2: Preliminary top-ten shares of routed IPv4 address mass under two identity mappings, redrawn at the precision of the imported manuscript's Figure 1 [33]. The exact snapshot date, effective mapped-mass denominators, and multiple-origin weighting policy remain unresolved; the difference is descriptive, not an isolated merging effect.

These preliminary values describe the imported comparison; they do not yet bound concentration over time. The source figure omits its exact snapshot date, the total address mass represented under each mapping, and the weighting policy for prefixes with multiple origins. We must reconcile mapping

coverage and those denominators before attributing the difference to merging alone. The shares measure routed IPv4 addresses, not traffic, customers, revenue, servers, or all allocated addresses. We also need to complete the manual mapping audit.

7.3 Identity models and measurement sensitivity

To determine how much a conclusion depends on identity assumptions, the analysis compares six mappings from ASN to entity while holding the routing, authorization, relationship, and customer-cone observations fixed. Table 7.1 describes each mapping; the sixth remains pending until the manual audit. The comparison tests sensitivity to plausible groupings. It does not rank the models from least to most accurate. CAIDA AS2Org and Borges offer alternative organization-level views, and agreement adds support for a particular merge. The planned audit-calibrated model will retain only merges that reviewers accept, rather than attempting a complete operator census.

The conservative-agreement model merges only pairs that both sources group together. This requires corroboration but leaves more possible siblings separate. The loose union merges pairs that either source groups together, covering more candidates at the cost of admitting relationships supported by only one source. We ask whether a result's direction, rank order, and magnitude remain stable across these choices. If they change, the disagreement shows how uncertain the operator-level conclusion remains even with a fixed BGP population.

Table 7.1: Identity models and their evidentiary meaning. No model establishes complete ownership or control.

Model	**Grouping rule**	**Supported interpretation**
ASN baseline	Each ASN is a separate entity.	Protocol-native baseline; no claim that entities are independent operators.
CAIDA AS2Org	Group ASNs using the time-aligned CAIDA organization mapping.	Administrative or organizational relatedness according to CAIDA's evidence and method.
Borges	Group ASNs using the Borges organization mapping.	A second inferred organization view with distinct source authority.
Conservative agreement	Merge only paired identities on which CAIDA and Borges agree.	High-confidence automated evidence of sibling structure, with lower coverage.
Loose union	Merge when either CAIDA or Borges groups the ASNs.	Broad sensitivity range; not a stronger control claim.
Audit-calibrated scenario (pending)	Retain accepted candidate clusters after reproducible manual review; make all other ASNs singletons.	Sensitivity under a restrictive accepted subset, not operator ground truth.

For the remaining manual audit, we will freeze the candidate list and have reviewers assess the candidates independently using a written evidence rubric. The audit-calibrated model will retain only accepted relationships and treat unreviewed, rejected, and ambiguous cases as separate ASNs. We will preserve the candidate list, dated evidence, reviewer labels, disagreements, and final adjudications so that another researcher can reproduce the accepted subset. Agreement between automated sources alone does not replace this review.

The routing corpus supplies the prefixes and origin ASNs to which we apply these mappings. It comes from CAIDA's RouteViews Prefix2AS snapshots [57] and currently contains 38,139,603 route-origin observations across 35 available dates from May 15, 2024, through May 5, 2026. For the

longitudinal analysis, the pipeline selects one snapshot per month from May 2024 through April 2026. It chooses the available IPv4 and IPv6 snapshot nearest the fifteenth, producing a 24-month panel. It expands multi-origin prefixes into one observation per origin ASN while retaining their common multiple-origin AS (MOAS) group. Keeping both the individual origins and their shared group lets the pipeline validate each origin and compare its identity while preserving the event in which several ASNs announced the same prefix.

The identity corpus combines 1,188,306 CAIDA AS-to-organization mappings from 10 source dates between April 1, 2024, and May 7, 2026, with 120,822 Borges organization mappings. After normalizing the six models, the current corpus contains 1,806,893 ASN-to-entity mappings across 11 source dates. For each routing snapshot, the pipeline uses the nearest prior identity mapping and never uses future identity information. This rule prevents a later acquisition, reorganization, or source update from changing how the analysis interprets an earlier routing observation.

To compare authorization across the mapped groups, we collected historical RPKI data for the AFRINIC, APNIC, ARIN, LACNIC, and RIPE NCC trust anchors. The collection contains 120 historical archive files, one for each of five trust anchors in each of 24 months; the parser read all files successfully. The normalized corpus contains 16,558,402 validated ROA payloads (VRPs) across 24 target dates. Prefix2AS and RPKI dates align exactly in the current run.

Standard route-origin validation [58] over 30,860,139 route-origin observations classified 17,301,136 as valid, 139,915 as invalid, and 13,419,088 as not found. These are ASN-level protocol results. The operator-level analysis then asks whether sibling ASNs have similar authorization coverage. Grouping the results does not change what makes a route valid, invalid, or not found under RPKI.

CAIDA AS-relationship snapshots and ASRank-style customer-cone data let us examine how grouping affects topology and rankings [28]. An AS-relationship edge connects routing domains; if both endpoints map to one entity, that edge no longer supports a claim about two independent operators. A customer cone describes the customer ASNs reachable downstream from an AS. Grouping sibling cones tests whether ranking ASNs separately splits one operator's apparent influence. To combine cones, the pipeline must count each shared descendant once. Summing member cones would count some downstream ASNs more than once.

For each identity model, the pipeline groups the routed resources by mapped entity and computes concentration over IPv4 prefix count, routed IPv4 address mass, IPv6 prefix count, and routed IPv6 address mass. The reported metrics include the shares held by the largest groups (top-5, top-10, and top-20), the Herfindahl–Hirschman Index and Gini coefficient, which summarize concentration and inequality, and effective diversity

$$N_{\text{eff}} = \frac{1}{\sum_i s_i^2},$$

where s_i is entity i's share of the measured resource. Effective diversity expresses the distribution as an equivalent number of equally sized entities.

Fixed inputs alone do not guarantee equal denominators: unmatched identities and the weighting of multiple-origin prefixes can change the address mass represented in a metric. In the remaining validation, we will retain unmatched cases explicitly and check that each comparison accounts for the same resource population. We will report coverage and weighting for each metric before interpreting a difference as the effect of grouping.

Additional pipeline passes aggregate route-origin validation by operator-month, count AS-relationship edges whose endpoints map to one entity, and reinterpret MOAS origins under each mapping. The pipeline also compares each prefix's origins in adjacent months and assigns one of four classes: same ASN, same-operator ASN change, different-operator change, or unmapped. These passes use the same observations across identity models, separating the effect of a mapping choice from a change in data collection.

7.4 Churn and supporting consequences of identity choices

Many observed origin-ASN changes remain within one mapped operator. Across the 24-month May 2024–April 2026 panel, 1,053,858 adjacent-month comparison events retain the same prefix but change its origin ASN. Figure 7.3 applies both mappings to those same events, including unmapped cases in the denominator. CAIDA associates 44.4% with the same operator, while the historical Borges subset associates 52.9%. The routing changes remain real; the mapping determines whether the analysis counts them as movement between operators. We use *churn* here to describe these changes between adjacent snapshots, rather than every routing change that may occur between them.

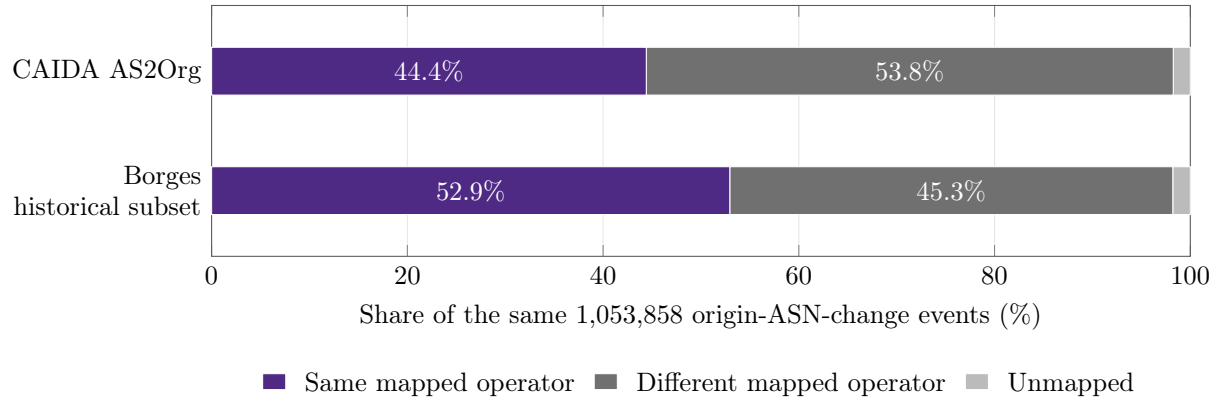


Figure 7.3: Preliminary interpretations of the same 1,053,858 origin-ASN-change events over 23 adjacent-month comparisons, May 2024–April 2026. Same, different, and unmapped counts are 468,073, 567,417, and 18,368 under CAIDA AS2Org, and 557,951, 477,035, and 18,872 under the historical Borges subset. Unmapped events remain in the denominator. These are comparison events, not unique prefixes or independently verified operator changes [33].

Both mappings use nearest-prior identity information. To construct the historical Borges subset, the study used dated AS2Org and PeeringDB inputs and omitted the AS detection, redirect, and favicon stages. This subset does not represent a full Borges campaign or independently establish common control. The analysis aggregates records of monthly changes by mapping and classification [33]; the prospectus artifacts preserve the source rows, hashes, and aggregates. The comparison therefore measures how those mappings affect the interpretation of churn, without identifying the operational cause of each change.

7.4.1 Authorization, topology, and customer-cone interpretation

The same identity choice also affects how an analyst interprets authorization coverage, network relationships, and downstream reach. These supporting analyses test consequences of grouping; longitudinal concentration remains Miscounting's primary question.

Mapped sibling ASNs can differ in whether they have RPKI authorization coverage. The current analysis finds 17,221 operator-month observations with mixed sibling VRP presence: at least one mapped sibling has coverage while another does not. The monthly count grows from 575 in the first panel month to 726 in the last. To obtain these counts, the pipeline groups each month's ASN-level authorization results by mapped operator. It preserves each route's original validation result rather than relabeling an ASN-valid route as operator-valid or vice versa.

This finding identifies uneven authorization coverage within mapped multi-ASN organizations. It does not show that an authorization is missing incorrectly or that all sibling prefixes should share one policy. RPKI supplies cryptographic authorization for a route origin; the identity mapping tells the analyst which ASN results to examine together. Neither their combination nor the observed mixture establishes that one team controls every ROA.

Some observed AS relationships connect routing domains within the same mapped organization. Applying CAIDA AS2Org to CAIDA AS-relationship edges places between 3,601 and 3,814 edges per monthly snapshot within one organization, approximately 0.6% of observed edges. Conservative agreement produces a similar fraction, while broader automated mappings classify roughly 0.7% as internal. These edges still describe relationships between routing domains; the grouping changes whether they support an interpretation as relationships between independent operators.

MOAS events expose the same distinction for prefix origins. Several ASNs originate a multi-origin prefix, but the identity mapping may associate those ASNs with one entity or with several. The analysis labels the former same-operator MOAS: several routing domains associated by that model. This label preserves the operational event while narrowing its interpretation. The current manuscript reports this distinction without determining the cause of each event or treating every same-operator MOAS as benign.

Customer-cone influence is the least mature analysis in this group. If one operator's downstream footprint is divided across siblings, ranking individual AS cones can understate its actor-level influence. The proposed operator-level measure combines the reachable customer ASNs across siblings, counting each once, and compares that union with the largest member ASN's cone. The difference, called hidden sibling cone mass, measures the additional reach that the largest individual cone misses. We still need to check overlapping cones, reconcile source definitions, and manually interpret prominent cases. The result remains provisional and does not serve as a principal empirical claim.

Table 7.2 summarizes the preliminary evidence and its remaining validation boundaries.

Table 7.2: Preliminary findings under explicit identity models [33].

Population	**Preliminary result**	**Supported claim and limit**
Imported final-snapshot routed IPv4 mass	Top-ten share: 26.4% CAIDA; 30.5% Borges.	Reported concentration depends on the mapping; effective denominators and weighting remain to be reconciled.
1,053,858 origin-ASN-change events	Same mapped operator: 44.4% CAIDA; 52.9% historical Borges subset.	Actor-level churn depends on the mapping; unmapped events remain in the denominator.
24 months of mapped sibling RPKI posture	17,221 operator-months have mixed VRP presence.	Sibling authorization surfaces can differ; no common policy or control is inferred.
Monthly CAIDA AS-relationship graphs	3,601–3,814 edges become internal under AS2Org.	Some AS edges do not support independent-operator edges; the routing observations remain valid.

7.5 What Miscounting establishes and what remains

The preliminary results show that identity mappings change reported concentration, apparent churn, and related interpretations of network actors. To determine which concentration trends hold over time, we still need to complete a reproducible mapping audit, align the dated inputs, account for the population represented in each denominator, and regenerate the longitudinal metrics. For the

customer-cone analysis, we must also validate how the pipeline constructs unions, handles overlap, and interprets prominent cases.

The accepted-merge scenario will provide a conditional lower bound on the correction of false independence. Each justified merge reunites ASNs of an accepted operator; leaving the other ASNs separate leaves some sibling relationships unresolved. This interpretation applies only to false independence. False unity, differences between ownership and operation, and model-specific coverage prevent it from bounding all identity error. Miscounting's intended contribution is to show which longitudinal conclusions remain stable under documented identity assumptions. Reproducible analysis, rather than preservation of a preliminary effect size, defines completion.

Even a defensible network-operator mapping leaves another question unanswered: *which actors perform the service roles within that network?* GOVT-MX addresses that question for government email.

Chapter 8

GOVT-MX: distinguishing government email roles

To identify an email dependency, an analyst must distinguish the organization running the mail service from the network hosting it. Consider the illustrative government domain `ministry.gov.example`. The government administers the domain; its MX record directs inbound mail to a gateway on a hosting provider's network, and its SPF policy authorizes a separate sending platform. Other tenants may share the gateway's ASN without sharing its administration. Treating the hosting network as the only service actor would create false unity. Figure 8.1 separates the roles that these records describe.

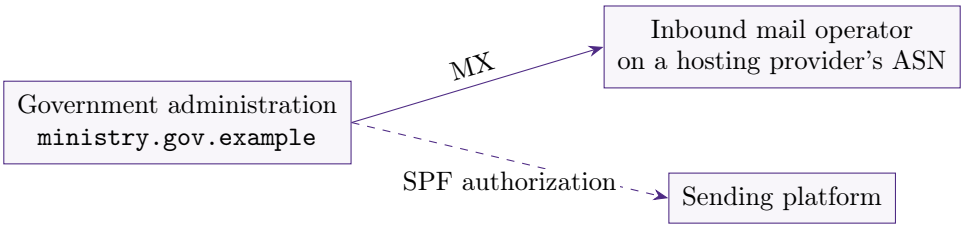


Figure 8.1: Schematic mail roles. MX identifies inbound handling and an address exposes network hosting; SPF authorizes sending without demonstrating that the platform sent mail. The actors need not differ in every deployment.

GOVT-MX asks who handles inbound government email and who has permission to send on the government's behalf. An MX record names hosts that accept inbound mail. Their resolved addresses identify the hosting networks, while SMTP banners and Transport Layer Security (TLS) certificates describe the service presented at those hosts. A gateway can conceal further processing or mailbox storage. SPF records supply a separate kind of evidence: they authorize systems to send on the domain's behalf without observing message flow [44]. The analysis therefore keeps domain administration, inbound handling, hosting, and outbound authorization separate.

The study uses a curated corpus of 11,875 federal-level, or equivalent, public-sector domains in 59 countries [30, 34]. The corpus combines public registries, official inventories, and manual verification, including state-owned enterprises in which the government holds a majority stake. Jurisdictions with accessible inventories are easier to cover, and the sample excludes subnational institutions. The corpus therefore represents the sampled institutions rather than a census of every public-sector domain.

We collected the measurements on April 22, 2026. Of the 11,875 domains, 9,107 (76.7%) published valid MX records and 2,768 (23.3%) did not. The MX-bearing population produced 30,890 MX rows, of which 30,582 resolved to an address, and 6,188 unique MX IP addresses across 1,398 origin ASes. To distinguish service administration from hosting and sending permission, the pipeline joins these DNS observations with the following evidence:

- IP-to-AS and AS-to-organization mappings [24, 25].
- Government-network [30], state-owned Internet service provider (ISP) [59], and National Research and Education Network (NREN) lists.
- SMTP extended hello (EHLO) banners and STARTTLS certificates.

- Mail Transfer Agent Strict Transport Security (MTA-STS) policies [60] and DNS-based Authentication of Named Entities (DANE) TLS Authentication (TLSA) records protected by DNS Security Extensions (DNSSEC) [61].
- SPF, Domain-based Message Authentication, Reporting, and Conformance (DMARC) [62], and observed DomainKeys Identified Mail (DKIM) [63].

These inputs describe visible mail infrastructure and published authorization at one measurement time. They do not record every message's path.

The pipeline classifies inbound service into four operating models. *Self-hosted* means that the available signals indicate a government agency, state-owned operator, or NREN provisions and administers the infrastructure. *Partial outsourcing* means a government-administered mail stack runs on commercial infrastructure. *Full outsourcing* means a commercial platform presents the visible mail service. *Potential multi-layering* means a secure email gateway is visible and other evidence indicates that someone has provisioned a different software-as-a-service tenant behind it. This last category remains conditional: a verification token can establish tenant provisioning without showing that mail currently reaches that backend.

These classes distinguish a government mail identity from government network hosting. For example, the measured MX hosts for `mfe.govt.nz` retain government names while their addresses originate in AWS AS16509. The evidence supports a government-administered mail identity on commercial infrastructure. The classes also separate inbound handling from outbound permission: after assigning the inbound class, GOVT-MX independently analyzes SPF authorization. A domain's inbound provider need not describe the systems it permits to send.

8.1 An evidence hierarchy for provider attribution

Different records can point to different organizations because they describe different roles. The classification pipeline resolves these apparent conflicts by asking which source can support the particular claim about visible mail operation. Table 8.1 gives the resulting hierarchy. For example, MTA-STS states the MX patterns that a domain owner authorizes, whereas BGP identifies the observed route origin. Using the former to classify mail operation preserves the latter's meaning as a routing observation.

Table 8.1: GOVT-MX evidence hierarchy [34]. Authority is limited to the mail role described in the final column.

Evidence tier	**Signals**	**Supported use**
Domain-owner authorization	MTA-STS MX patterns and DNSSEC-protected DANE/TLSA records	Explicit authorization of inbound MX identities by the measured domain.
Server-presented identity	STARTTLS certificate names, SMTP EHLO banners, and MX hostnames	Provider or administrative identity presented by the visible inbound service.
Network and organization	Origin ASN, AS-to-organization mapping, NREN, state-owned-ISP, and government-network lists	Network placement and evidence that a government-associated network carries the endpoint.
Auxiliary role evidence	SPF includes and apex provider-verification tokens	Outbound sending authorization, or potential backend provisioning; neither proves observed message flow.

The pipeline first checks whether the domain owner explicitly authorizes an inbound service through MTA-STS or DANE/TLSA. An enforced MTA-STS policy, fetched over HTTPS, lists the permitted MX patterns. A TLSA record protected by DNSSEC ties a permitted MX service to key or certificate material. These records provide stronger evidence of domain-owner intent than a service name inferred from network ownership. They cover only part of the sample: 401 domains (3.4% of the full corpus) published MTA-STS and 640 (5.4%) published DANE/TLSA. An absent policy leaves the identity unresolved at this stage; it does not show that a provider is absent.

When domain-owner authorization does not resolve the class, the pipeline examines the identity that the reachable server presents. TLS certificate common names and subject alternative names, EHLO banners, and provider-specific MX namespaces can identify a commercial mail platform on shared infrastructure. These signals describe an operational identity, not legal ownership: customers can configure names, and certificates or banners may be stale or generic.

If those signals also leave the class unresolved, the pipeline uses network evidence. It maps each MX address to its origin ASN and organization, then checks government-network, state-owned-ISP, and NREN lists. This step prevents the classification from automatically treating a state-owned telecom or NREN as commercial outsourcing. The pipeline classified only 2,441 MX rows (7.9%) from network ownership alone.

Two separate passes identify roles outside this inbound hierarchy. One flags a secure email gateway combined with a provider verification token as potentially multi-layered. The token establishes provisioning, so the pipeline does not count the indicated backend as a confirmed message processor. The other follows SPF include relationships and maps the referenced authorization to commercial providers. This identifies domains that permit an external sender even when the inbound service appears self-hosted.

The study measures DKIM deployment separately as part of email security posture. To find a DKIM key in DNS, the measurement must know its selector, the name that identifies the relevant key record [63]. DNS does not enumerate selectors. Probing common and provider-specific selectors therefore supplies an observed lower bound on deployment, rather than a complete count.

For MX-hostname domains that remained unresolved, we used a constrained language model to propose provider-specific hostname rules. The model received only hostname evidence and could select only from a manually curated set of 60 mail providers. It proposed 14 rules; manual reviewers retained eight unambiguous provider-controlled namespaces. The accepted rules classified 1,382 previously unresolved MX rows, 4.5% of all MX rows, with no conflicts against the manual baseline. The model proposed reusable rules, while human reviewers decided which rules the pipeline could apply.

The hierarchy changes attribution in practice. Higher-tier evidence overturned a network-only assignment for 4,771 conflicted MX rows. The pipeline also flagged 4,250 rows associated with security appliances that may front another provider. Retaining these cases shows where an address-to-AS lookup can correctly identify a routing origin yet assign the wrong application role.

8.2 Control, placement, and outbound delegation

Third parties appear in the visible inbound infrastructure of 59.9% of the 9,107 MX-bearing, host-classified domains: 48.8% are fully outsourced, 1.8% are partially outsourced, and 9.3% are potentially multi-layered. The pipeline classified the remaining 40.1% as self-hosted under the available signals (Figure 8.2(a)). That share is an upper bound on government-run inbound handling because a visible gateway can conceal additional processing.

Self-hosted inbound service can coexist with permission for a commercial provider to send. Among

3,652 domains classified as self-hosted, 1,112 (30.4%) authorize at least one commercial outbound sender through SPF (Figure 8.2(b)); Microsoft is the most common such authorization. This finding describes permission, not whether an authorized provider sent a message during the measurement window. It shows why an inbound self-hosting label cannot establish exclusive government control over all mail roles.

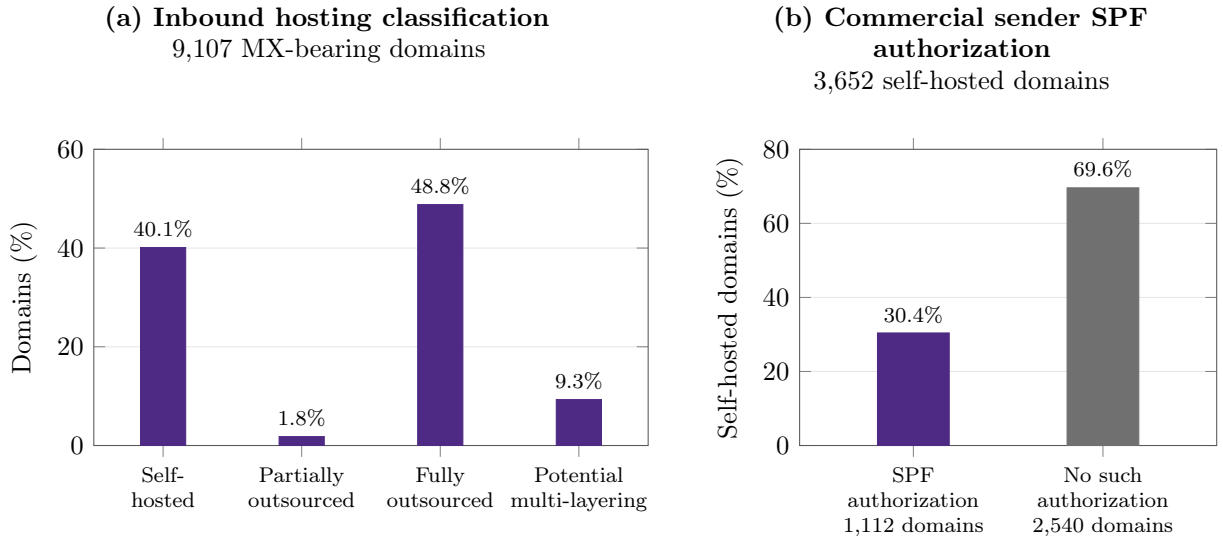


Figure 8.2: Inbound hosting and outbound authorization describe different mail roles in the April 22, 2026 snapshot [34]. (a) Hosting classes among 9,107 MX-bearing, host-classified domains. Potential multi-layering indicates gateway and provisioning evidence, not confirmed backend handling; self-hosting remains an upper bound on government-run inbound handling. (b) Among 3,652 self-hosted domains, 1,112 (30.4%) authorize a commercial sender through SPF and 2,540 (69.6%) have no such authorization. SPF authorization does not establish observed sending.

To evaluate the classifications, we compared the pipeline with manual labels for 400 MX-bearing domains. We drew 200 domains uniformly and 200 from higher-risk strata: network-only, partial-outsourcing, gateway, unknown, conflict, and language-model-affected cases. Reviewers labeled the domains without seeing the pipeline output. The binary outsourcing decision agreed with their labels on 192 of 200 random domains (96.0%) and 193 of 200 stress-sample domains (96.5%). The stricter operational class agreed on 365 of 400 domains (91.2%). A separate 50-domain SPF audit recovered the normalized outbound-provider set for 47 domains (94.0%) [34]. These checks support the measured decisions on the selected samples, while leaving uncertainty about individual provider assignments and cases with sparse evidence.

Provider use is concentrated but not uniform. GOVT-MX identifies 41 non-government providers. Microsoft supplies 54.4% of outsourced domains and Google 11.9%; over all 9,107 domains with MX records, their respective shares are 28.0% and 6.1%. Only those two providers appear in at least 25 countries, while 17 providers appear in a single country. This result establishes concentration in the measured visible mail roles. It does not establish substitutability, contract concentration, or common failure behavior, and the provider counts depend on the attribution hierarchy rather than on ASN counts alone.

The apparent foreign placement of mail endpoints depends strongly on how the analysis handles shared infrastructure. The raw IP-geolocation view places an endpoint outside the domain owner's country for 34.7% of MX-bearing domains. Shared anycast or multi-tenant front doors can inflate

this share because an address database may return a provider location rather than the site serving a particular tenant. Restricting the analysis to the manuscript's reliable-geolocation subset reduces the share to 13.9%. Both percentages describe visible MX endpoints under the stated filters; neither locates mailbox storage, data processing, or legal jurisdiction.

The European Union subset shows the same sensitivity. Among 3,092 geolocatable EU domains, the raw view places 63.4% domestically, 30.9% elsewhere in the EU, and 5.8% outside it. After reducing shared-platform ingress effects, the corresponding reliable-subset shares are 90.8%, 7.8%, and 1.4%. The change shows how the treatment of shared endpoints affects the location result. It does not establish that either view recovers tenant residency; that stronger claim requires provider or contractual evidence.

Email security indicators also differ across hosting classes, but the pattern is mixed. In the selected probing set, fully outsourced domains have more observed DKIM than self-hosted domains, 72.3% versus 36.2%, and more MTA-STS, 6.3% versus 1.1%. Terminal SPF `-all` appears at similar rates, 62.6% for fully outsourced and 58.6% for self-hosted domains, as does DMARC enforcement, at 51.9% and 52.7%. These associations come from one snapshot, with incomplete DKIM selector discovery and possible differences in country policies. They do not establish that a hosting model causes stronger email security.

8.3 Scope and contribution

GOVT-MX distinguishes visible inbound operation, network hosting, and outbound sending authorization in the sampled government domains. Its coverage and attribution limits determine how readers can use those results. The curated corpus covers countries unevenly, and small-provider or hybrid deployments can remain ambiguous. Self-hosting therefore remains an upper bound on government-run inbound handling. MX geolocation describes endpoints rather than mailbox residency, data processing, ownership, or jurisdiction. SPF records authorize sending, provider tokens indicate provisioning, and DKIM probing finds only discovered selectors; these signals do not record message flow. The single snapshot also cannot establish trends or causal effects of hosting class on security posture.

The two studies answer complementary identity questions. Miscounting asks which ASNs an analyst should group when estimating concentration and change. GOVT-MX asks which service roles the analyst must keep distinct even within one mapped network. In both cases, making the choice explicit lets readers see how the observations support an actor-level conclusion and where that conclusion depends on an inferred grouping or role. The dissertation will use this distinction to connect operator sensitivity and mail-role attribution without treating routing, presented identities, and authorization as interchangeable evidence.

Part III

Protocol extensions

DNS endpoints reveal only part of the infrastructure that handles a query. This part measures that visibility gap and develops protocol extensions through which cooperating resolvers can report participation and clients can authenticate the reports they receive.

Chapter 9

Measuring DNS resolver endpoints

Consider a user who selects a recursive DNS resolver because of its privacy policy. The user can see the address to which the client sends a query and the answer that comes back. The selected resolver may nevertheless forward the query to another resolver, which may in turn rely on additional infrastructure. Those participants can handle the query without appearing in the answer. Knowing which resolver the user selected therefore leaves an operational question: *who else handled this query?*

This chapter begins to address the dissertation's fourth research question by measuring which DNS participants are visible to the endpoints. We compare the resolver contacted by a client with the source address recorded by an authoritative server under our control. This comparison identifies differences between the two visible endpoints. Ordinary DNS responses, however, do not enumerate the participants between them.

The missing sequence differs from an ambiguous PTR token or ASN-to-entity mapping: it is not encoded in the observed endpoints. Refining an inference model alone cannot recover a participant that leaves no such signal. DNSTRACING first measures this visibility gap, then supplies a cooperative reporting channel and a mechanism for checking the reports received by the client.

This part develops DNSTRACING in three stages [42]:

1. The measurement compares the client-facing and authoritative-facing resolver endpoints. A difference establishes an endpoint mismatch; it does not recover a path.
2. A protocol extension lets cooperating resolvers report their participation. Their entries expose participants that endpoint measurements omit, but the sequence remains a report.
3. Response-bound attestations let the client check that received entries refer to its query and the selected response. The client verifies signatures, intact hash-chain order, nonces, and timestamps. Non-participating hops remain outside the evidence.

The two chapters therefore move from observation, to disclosure, to verification without treating any stage as complete path discovery.

9.1 Constructing endpoint observations

To determine whether the client and authoritative server see the same resolver, we use RIPE Atlas probes [47] as clients and a domain under our authoritative control. For each query, Atlas records the resolver address contacted by the probe. DNSTRACING calls this address the *ingress*, or client-facing *frontend*, resolver. The authoritative server records the source address of the system that performs the final lookup and returns that address in its response; this is the *egress* resolver. The analysis joins records by query into the triple (C, I, E): client C, frontend I, and egress E.

The triple (C, I, E) records endpoints rather than a complete resolver path. It leaves intermediate forwarders, forwarding depth, fan-out, cache-dependent changes, and multiple upstream queries for one client request unobserved. When I and E differ, the client contacted a different endpoint from the one visible to the authoritative server. Forwarding could explain the difference, but the endpoint

records alone cannot identify the intervening mechanism. Chapter 10 develops a protocol extension to supply this missing participation evidence.

The analysis must handle local addresses explicitly because a private frontend address has no globally meaningful ASN mapping. The full-population AS analysis conservatively assigns such frontends to the client AS. Campaign B contains 6,713 private-ingress observations, or 40.4% of its 16,636 successful observations. The analysis also counts the 83 rows without a complete client/frontend ASN pair as same-AS. Both conventions reduce apparent client-to-frontend boundary crossing. A private address identifies a near-side resolver endpoint without establishing that the ISP operates it or that its upstream remains in the same AS.

To compare network and organizational boundaries, the analysis maps public client, frontend, and egress addresses to origin ASNs using RouteViews data [64] and to candidate organizations using Borges [27]. As in Chapter 7, an ASN mismatch describes a routing boundary. An organization mismatch depends on the mapping and does not prove independent control or ownership.

Country comparisons require different rules for anycast and unicast addresses. One anycast address can represent many serving sites, so an address-level database entry may not identify the site that handled a query. For known public anycast frontends, the analysis first uses operator-reported NSID or diagnostic TXT site identifiers from Google Public DNS, Cloudflare, Quad9, and OpenDNS. It next uses location-bearing PTR names interpreted with The Aleph [21]. If neither source identifies a site, the probe–resolver pair remains unresolved in the geolocatable-anycast analysis. These labels report serving sites; they are not independent physical attestations. The primary egress-country analysis excludes anycast egress addresses.

For other public addresses, IPinfo [65] supplies a fallback location. RIPE Atlas round-trip time (RTT) and traceroute constraints test the database-derived unicast labels. The full-population analysis conservatively counts contradicted labels as domestic, while a stricter plausibility-consistent subset excludes them. Reporting both populations makes the effect of the geolocation rule visible.

We conducted four campaigns from 2023 through 2025. Between 10,082 and 11,918 unique probes returned results per year across 187–190 countries and 3,100–3,150 ISPs. Campaign-level ASN-completeness filters retained 69,480 observations across the four campaigns; Campaign B supplies the 16,636-row 2025 cross-sectional base. In the 2025 dataset, the client population contains 11,444 unique IP addresses in 3,805 ASes. The frontend population contains 5,538 addresses in 2,294 ASes, and the egress population contains 7,563 addresses in 1,565 ASes. RIPE Atlas is weighted toward fixed-broadband and institutional access networks. These counts describe the measured population, not a random sample of Internet users.

9.2 Boundary crossings and concentration

The client-facing and authoritative-facing endpoints frequently differ in the measured population. Figure 9.1 shows that frontend–egress divergence remains common when the comparison moves from IP addresses to IPv4 `/24` prefixes in the public-ingress population. Country labels reveal another endpoint difference: among 15,125 triples with a non-anycast egress and a known egress country, 2,826 (18.7%) place the egress outside the client's country. These comparisons establish where the observed endpoints differ; they leave the intervening sequence unresolved.

AS-boundary crossings remain common under the conservative treatment of private frontends (Table 9.1). Under the full-population convention, 6,622 of 16,636 observations (39.8%) place client and frontend in different ASes. Restricting the population to public ingress raises the observed share to 6,480 of 9,877 (65.6%); the corresponding mapped-organization result is 6,206 of 9,773

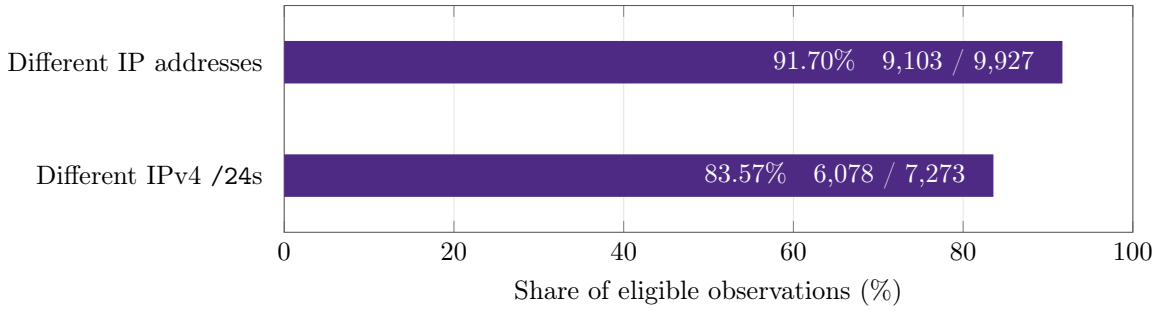


Figure 9.1: Frontend–egress divergence in Campaign B public-ingress observations [42]. The IP comparison requires both endpoint addresses; the `/24` comparison additionally requires both to be IPv4. Counts and denominators are shown separately. Different endpoints do not establish an intervening forwarding hop or its depth.

(63.5%). These are distinct views of endpoint divergence, not estimates of the number of unobserved intermediaries.

Table 9.1: Selected DNSTRACING endpoint results [42]. Each percentage applies only to the population named in the first column.

Population	**Result**	**Bounded interpretation**
16,636 Campaign B observations	6,622 (39.8%) have client ASN ≠ frontend ASN under the conservative convention.	Lower bound on observed AS-boundary crossing; unattributable ingress is not assigned an owner.
9,877 public-ingress observations	6,480 (65.6%) have client ASN ≠ frontend ASN.	Publicly attributable subset; 6,206 of 9,773 classified observations (63.5%) also cross mapped organizations.
5,498 scheduled anycast frontend pairs	989 (18.0%) are observed outside the client country when unresolved pairs count as domestic.	Conservative rate for the scheduled anycast population, not all resolver types.
3,869 geolocatable anycast frontend pairs	989 (25.6%) are observed outside the client country; Google supplies 67.3% of those 989 cases.	Rate and provider mix for identity-bearing anycast pairs; unresolved pairs are excluded.
15,125 non-anycast-egress triples with known egress country	2,826 (18.7%) have client country ≠ egress country.	Endpoint-country mismatch after excluding anycast egress; not a complete path-country trace.

The anycast country result depends on which pairs have usable site labels. Among 3,869 anycast probe–resolver pairs with a usable operator identity, 989 (25.6%) are served from a country different from the probe. Counting all 1,629 unresolved pairs as domestic makes the same 989 events represent 18.0% of 5,498 scheduled pairs. The interval 18.0–25.6% therefore brackets the observed rate within the anycast subset under these two denominator choices. It is not a lower and upper bound for all recursive DNS traffic.

Google Public DNS accounts for most cross-border anycast pairs: 67.3% of the 989 out-of-country pairs, compared with Cloudflare's 20.6% and Quad9's 11.4%. Google's share of all 3,869 geolocatable anycast pairs is 52.66%, so 67.3% must not be read as its share of all measured DNS

resolution. Among the 989 cross-border pairs, 570 (57.6%) retain egress in the same foreign country as the frontend. Another 264 (26.7%) use an egress in a third country, and 155 (15.7%) lack an egress-country label.

The egress-country comparison also includes queries through frontends outside the public anycast subset. Among 15,125 observed triples with a non-anycast egress and a known egress country, 2,826 (18.7%) place egress outside the client's country. Over all 16,563 triples with valid client, frontend, and egress country fields, including the cases excluded from the narrower egress population, the client–egress mismatch rate is 17.06%. Again, this is an endpoint comparison. A query may have crossed another country between the two observed endpoints without being counted, or may have reached an anycast site whose label is inaccurate.

One frontend can correspond to several observed egress endpoints. Among groups defined by a probe and frontend IP, 40.42% map to more than one egress IP and 6.73% to more than one egress ASN. These changes may result from load balancing, cache state, forwarding policy, or other mechanisms. They support the claim that a single client-visible resolver address does not determine one authoritative-facing endpoint; they do not identify the hidden topology.

Out-of-AS endpoint placement is also associated with higher measured latency. For frontends, median Internet Control Message Protocol (ICMP) RTT is 2.20 ms when the resolver is in the client AS and 6.84 ms when it is outside. Median cached-DNS time is 2.80 ms and 8.95 ms, respectively. For egress, the corresponding medians are 3.67 ms and 11.59 ms for ICMP and 6.28 ms and 12.13 ms for cached DNS. The same association appears across the measured roles and methods. The comparison does not establish causality, and AS mismatch does not measure geographic distance.

Within a stable panel, multi-AS triples became more common over time. To reduce the effect of changes in Atlas composition, the longitudinal analysis retains 2,468 probes present in all three year-level datasets whose client AS and country remain fixed. Under probe-weighted aggregation, triples whose three ASNs match ($C = I = E$ in this comparison) decline from 41.5% in 2023 to 37.9% in 2025. Client-to-frontend ASN mismatch rises from 35.5% to 42.1%, frontend-to-egress ASN mismatch rises from 39.6% to 43.9%, and triples with both third-party frontend and different-AS egress rise from 16.5% to 23.9%. The middle year is not monotonic in every category. The supported conclusion is a directional increase in multi-AS observed triples within this stable panel, not a universal trend in DNS outsourcing.

9.3 Limits of endpoint measurement

The endpoint measurement has four principal limitations:

- Atlas probe deployment overrepresents European fixed-broadband and institutional networks. Observations that share an AS or resolver configuration are not independent.
- The analysis assigns private frontends to the client AS to avoid overstating boundary crossing, but it does not verify that assignment.
- The experiment observes only the frontend and egress. It cannot distinguish a forwarder from load balancing, anycast catchment, or other internal architecture.
- The location sources support different claims. Operator identity strings report anycast serving sites, while PTR and database coverage is incomplete. Active checks can reject an implausible country without proving a remaining label correct.

  Every reported rate must therefore retain its endpoint population and geolocation rule.

9.4 What the endpoints leave unreported

Two different resolver arrangements can produce the same observed triple (C, I, E) (Figure 9.2). The frontend might contact the egress directly, or it might forward through other resolvers before the same egress reaches the authoritative server. The experiment distinguishes neither arrangement from the endpoint records alone. Even matching frontend and egress identities would not establish that no other system handled the query. The endpoint comparison answers which systems were visible at the two ends. Answering who else participated requires additional evidence.

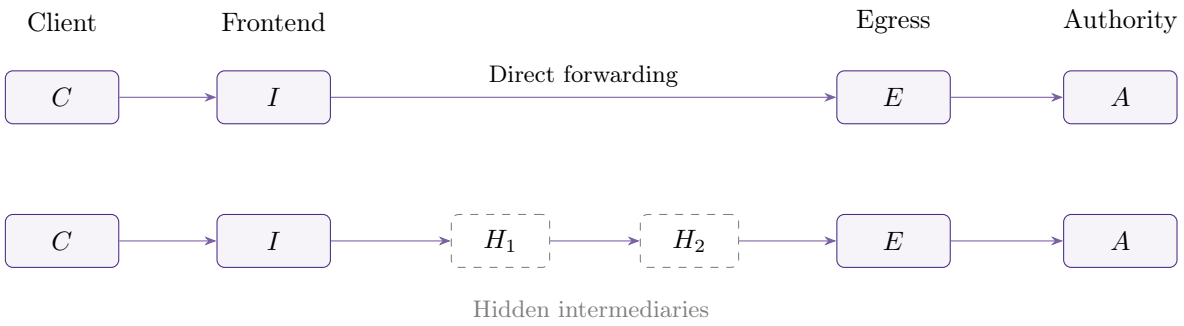


Figure 9.2: Illustrative arrangements with the same observed client, frontend, and egress. Dashed nodes leave no entry in the endpoint measurement; the measurement does not identify either hidden sequence.

Chapter 10

Disclosing and verifying DNS resolver participation

Endpoint observations leave intermediary participation unresolved. This chapter develops cooperative reporting and response-bound attestation, then evaluates what the reports establish and what they cost in controlled deployments [42].

10.1 Cooperative resolver-path disclosure

To expose participants between the visible endpoints, resolvers need a way to pass their reports to the client. Existing DNS mechanisms identify individual servers without accumulating a sequence of participants. NSID lets a responding server return an operator-defined identifier, and some resolvers expose related CHAOS TXT identities; both describe one responding system [39]. DNSTRACING prototypes two ways for resolvers to disclose their participation. Each depends on cooperation: an intermediary must implement the mechanism or at least preserve its metadata. Neither design discovers a component that silently omits itself.

10.1.1 Iterative query tracing

Technique 1 lets the client ask each resolver which resolver it would contact next. The client sends the dedicated experimental query type `TYPE65400` to its frontend, which returns its identity and the address of the next resolver. The client queries that next hop and continues until reaching the authoritative server. This step-by-step walk lets the client associate delay with each reported segment and retain partial results if a later resolver does not participate.

The walk is an on-demand diagnostic that requires direct client access to each reported next hop. A forwarding resolver may reject these queries, apply an access-control list, or report an unreachable address. Providing the next hop's IP address also exposes information an operator may not want to publish. The prototype completed iterative discovery in the tested chains; reachability and disclosure policies on the public Internet remain untested.

10.1.2 In-band EDNS tracing

Technique 2 carries participant reports within an ordinary query and response, using experimental EDNS option `65024`. The client sends an empty resolver-path option with its query. Each participating resolver appends its identifier before forwarding the query, and the authoritative server copies the accumulated option into its response. The client thus receives the reported sequence in the original round trip. EDNS provides the message extension [38], but deployed recursive resolvers need not propagate an unfamiliar option. The current results cover resolvers with explicit support and the tested cases in which intermediaries preserve the option.

The two techniques report different sequences. Iterative tracing follows the next hops reported during a diagnostic walk. In-band tracing reports participation associated with an ordinary lookup and its returned response. Cache state, forwarding policy, and response selection can differ between those requests, so a diagnostic walk need not reproduce the ordinary-response path. Both techniques add participant reports; the next step is to check the origin and integrity of those reports.

10.2 Verifying reported path participation

The client needs to check who signed each received report, whether anyone changed it, and whether it refers to the query and response at hand. DNSTRACING implements these checks with Ed25519 signatures [66] linked by hashes. The resulting *response-bound attestation* ties each signed entry to the response that the resolver selected, so a valid report for one candidate answer cannot be substituted for a different answer or status without detection.

A response producer creates a fresh response identifier only after observing the response it will return. Each participating resolver verifies the chain attached to its selected downstream response, then signs its claimed ASN and country together with the client nonce, canonical question, canonical response digest, timestamp, response identifier, and preceding-entry hash. The nonce identifies the client's query attempt; the canonical question and response digest provide consistent representations for checking the question and answer. The preceding-entry hash links each entry to the chain it extends. The client verifier checks these fields, the signatures, and the hash links.

The design specifies how resolvers handle cache hits, retries, fan-out, partial participation, and resource limits. When answering from cache, a resolver creates fresh metadata for the current nonce instead of reusing the chain from an earlier request. When it receives an unattested response, a participating resolver can start a chain marked `GAP_BELOW`. It must leave malformed downstream data invalid rather than turn it into a valid partial report. When retries or fan-out produce competing responses, the resolver keeps their identifiers and chains separate and extends only the response it selects. The client then distinguishes valid received evidence, invalid evidence, an unattested response, and an unknown or revoked key.

Each signed entry asserts an ASN–country pair. ASN alone is too coarse for the measured location question: among 16,141 plausibility-consistent observed triples, 3,166 cross a country boundary, and 1,091 of those 3,166 (34.46%) retain the same ASN across the relevant country labels. The implementation uses an RPKI Resolver-Path Key Authorization object to authorize an Ed25519 key for the ASN under a private test trust anchor [67]. The validator checks the trust chain, manifest and certificate-revocation list, certificate resource coverage, validity interval, and key identifier. These checks establish the ASN-to-key authorization; RPKI does not certify the signed country field. Standardizing the object type and operating production publication infrastructure remain deployment work.

Attestation authenticates the entries the client receives; it cannot establish which resolvers are absent. A resolver cannot forge another operator's entry without its private key, and it cannot alter received signed bytes or intact hash links without detection. It can omit its own entry, and a non-participating resolver can process the query without identifying itself. Arbitrary suffix, whole-option, or replacement stripping is not generally detectable. A valid report therefore authenticates the received cooperative assertions, their order, and freshness under the evaluated assumptions. It does not establish assertion truth, physical traversal, legal jurisdiction, path completeness, or the absence of hidden resolvers.

10.3 Controlled implementation and evaluation

The completed evaluation asks two questions: do the reports match the controlled events they describe, and what does generating and checking them cost? We tested diagnostic walks, selected-response paths, and received signed assertions separately because each represents a different kind of evidence.

10.3.1 Reported paths and unsigned disclosure

We implemented both reporting mechanisms in BIND 9, Unbound, and Knot Resolver. To measure their performance, we replayed a ten-minute university DNS trace containing 2,616,894 queries for 106,102 names against controlled resolver chains. We anonymized query-source identifiers and query names before replay; a local authoritative server answered every replay name. The cloud experiments used AWS `c7i.large` instances with two virtual CPUs (vCPUs) and 4 GiB of memory. They varied offered load from 2,000 to 20,000 queries per second (QPS) and compared unmodified BIND, modified BIND with tracing disabled, and the two reporting techniques. Separate Docker experiments varied chain depth, cache state, and resolver implementation.

The reported paths matched the controlled resolver events. Here, a response path is the sequence of participating resolvers whose processing contributes to one selected DNS response. Across 185 controlled runs covering 23 scenarios and 32 topologies, validators matched all 63,042 individual paths and 47,042 selected-response paths to linked per-resolver events. The validators marked all 1,000 intentionally partial responses incomplete, and no sibling- or cross-response merge occurred. Controlled cache tests likewise matched all 17,000 selected-response paths to current-query events without reusing the path that populated the cached resource-record set.

At the common 20,000-QPS offered load, Technique 2 changed median throughput by -0.90% and median 95th-percentile latency by $+2.08\%$ relative to modified BIND with tracing disabled. The largest measured central processing unit (CPU) increase was 17% in one cross-region workload; memory stayed within 2 MB of vanilla BIND. In the local depth experiment, reported-path recovery was 100% through depth 15, with no observed truncation or Transmission Control Protocol (TCP) fallback. The 95th-percentile latency rose from 0.077 to 0.097 ms at depth 15. Four depth-three BIND, Unbound, and Knot combinations also recovered every configured path in the controlled testbed.

Technique 1 completed every diagnostic walk in the evaluated cloud and Docker configurations. However, the diagnostic and ordinary-response paths differed in 4,000 of 7,000 paired trials. This result confirms the need to evaluate the two sequences separately.

In the local depth experiment, all 200 walks completed at each tested depth from two through 15. At depth 15, end-to-end discovery took 4.015 ms at the median and 5.158 ms at the 95th percentile. In geographically distributed cloud chains, delay reflected the network path rather than protocol processing. The median total ranged from 2.117 ms for a domestic topology to 61.174 ms when the path extended across three regions. Technique 1 therefore exposes useful per-hop timing in a controlled deployment, but each additional reported hop also requires another client query.

These experiments establish implementation feasibility and overhead for the specified hardware, workloads, and topologies. The maximum offered load of 20,000 QPS is not a measured production capacity ceiling. Deployment also depends on intermediaries retaining the extension, regardless of its processing cost. Measuring option survival and behavior across deployed resolvers remains part of the proposed work.

10.3.2 Attestation conformance, wire behavior, and cost

Standalone Go and Python implementations agreed on the published specification's test cases. To check that separate code paths interpret the same messages consistently, each codebase processed 18 canonical vectors for five repetitions without importing the production implementation or vector generator. Across 180 implementation decisions, the codebases agreed on 90 of 90 paired verdicts and 85 of 85 signer-applicable canonical signatures. All 180 decisions matched the expected verifier

state, parser and serializer bytes, and canonical response hash. Each decision links to a client event, separate implementation log, and one of 180 packets in five packet captures.

The mutation cases test whether the verifier responds correctly when entries are removed, reordered, or modified; signatures are copied; an old cache chain or nonce is reused; or an answer, status, or sibling response is substituted. They also cover partial participation, stripping, key state, and overflow. This evaluation establishes agreement between separate code paths for the published vectors. Production interoperability remains to be evaluated.

We checked message size and transport behavior in 240 runs across 48 conditions, reconciling 1,840 packet captures with 1,600 linked event sets. The option uses a 36-byte header and 144 bytes per signed entry, or $40 + 144h$ bytes including the EDNS option-code and length fields, for at most six entries. Once a chain reaches that limit, an upstream participant forwards it without appending, and the verifier marks the result resource-limited. At a 1,232-byte UDP payload, captures confirmed the specified truncation and TCP fallback for responses that exceeded the size limit while preserving the available chain. Signed depth-eight and depth-15 cases used TCP in 10 of 55 and 45 of 125 cases per IP family, respectively; their reported chains remain subject to the entry limit.

Controlled microbenchmarks measured median per-entry signing cost between 36.8 and 37.1 microseconds and median verification cost between 93.4 and 95.2 microseconds. To evaluate how these costs affect query completion, we ran a 450-run, 37.5-hour depth/load matrix with depths two, three, and five, three modes, ten offered loads through 20,000 QPS, and five repetitions per cell. Figure 10.1 compares signed completion with the reported baseline at 20,000 offered QPS across the three depths; signed loss stayed below 0.001% across all loads. Completed heterogeneous BIND, Unbound, and Knot runs achieved a verification rate of 1.0 with no observed truncation or TCP fallback. These controlled results establish implementation, conformance, wire, and performance properties for the tested systems; they are not production capacity or Internet-deployment guarantees.

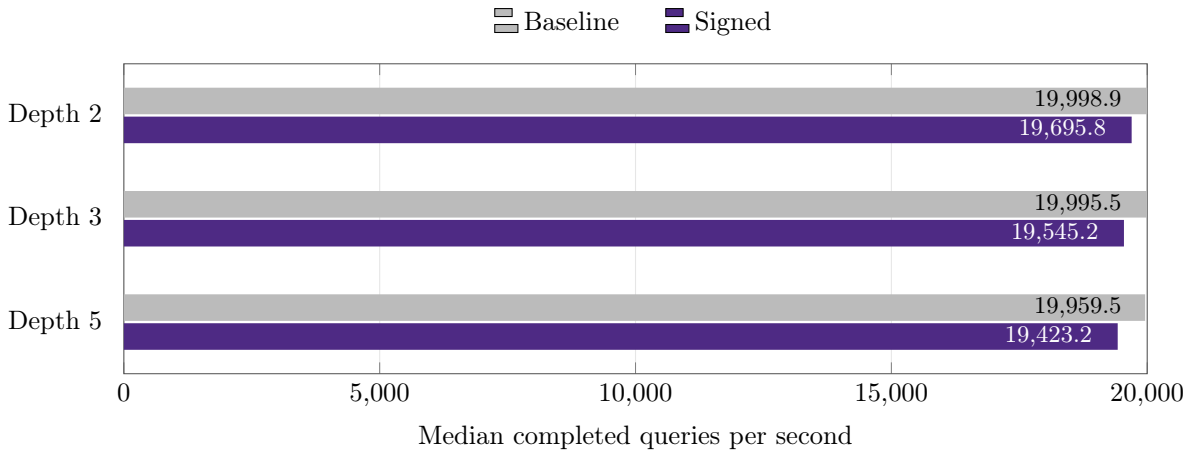


Figure 10.1: Controlled Resolver-Path throughput at 20,000 offered QPS [42]. Bars show medians over five 300-second repetitions per depth and mode; labels round the published medians to one decimal. “Baseline” preserves the source’s comparator label. The reported medians do not provide run-level uncertainty or establish production capacity.

10.4 Evidence limits and remaining deployment work

Together with Chapter 9, this chapter gives an auditor three kinds of evidence: observed endpoints, participant statements, or statements whose origin and integrity the client can check. Table 10.1 summarizes the claim each stage supports. None establishes a complete request path.

Table 10.1: Evidence and claim boundaries in DNSTRACING.

Evidence	**Auditable claim**	**Not established**
Controlled endpoint observation	Which frontend the probe contacted and which egress address reached the authoritative server.	Intermediate resolvers, forwarding depth, or the mechanism causing a mismatch.
Cooperative resolver report	Which participating implementations report themselves in the returned sequence.	Honesty, completeness, or option survival through arbitrary intermediaries.
Response-bound hash-chained attestation	Signature validity, received-entry order and integrity, query and selected-response binding, and freshness under the key assumptions.	Assertion truth, physical traversal, absence of omitted participants, path completeness, or confidentiality.

The controlled protocol evaluation covered three resolver implementations, cache hits, retries, fan-out, heterogeneous chains, and intentional partial participation. Production DNS adds resolver stacks, middleboxes, operator policies, access controls, and workloads beyond those tests.

Technique 1 may fail when a downstream resolver will not answer the client. Technique 2 may fail when an intermediary strips or reconstructs EDNS options. The study has not yet measured public-Internet option survival, an independently administered production deployment, or broader production interoperability. The current evidence supports controlled feasibility and conformance, not deployability at Internet scale.

Operating attestation also requires key management and resources to reject replayed responses. The prototype evaluated a 300-second freshness window and verifier nonce state. Production-scale nonce retention, publication of the Resolver-Path Key Authorization (RPKA) object, key rollover, and operational revocation remain open. Per-hop signatures consume CPU and message space. AS–country labels sacrifice site precision, while finer labels expose more of the topology. The completed Go/Python conformance test provides a baseline for checking the published vectors; an operational verification service requires the remaining deployment work.

DNSTRACING connects the endpoint visibility gap to a way of obtaining and checking additional participation evidence. Measurement identifies the observed frontend and egress. Cooperative disclosure adds participant reports, and attestation lets the client check their origin, order, freshness, and binding to a selected response. The completed measurements and implementation tests establish those capabilities within the reported populations and testbeds. Public-Internet option survival, production-like key lifecycle behavior, and an independently administered deployment attempt remain the empirical work needed to assess deployment. Throughout, silent intermediaries remain outside the claim.

Dissertation plan and synthesis

The remaining chapters bring the three research parts together in a common plan for completing and evaluating the dissertation. They specify the remaining studies, expected contributions, and target milestones while preserving each study's evidentiary limits.

Chapter 11

Proposed work

Three empirical studies remain: GeoCorrect's full-day and month-long study, Miscounting's mapping audit and longitudinal analysis, and Resolver-Path's deployment evaluation. Parts I–III establish the questions, completed evidence, and implemented methods. This chapter explains how we will conduct those studies and what records they will produce; Chapter 12 defines their evaluation and completion criteria. Dissertation integration and Bet provenance work are separate from these experiments.

11.1 GeoCorrect: full-day and month-long analysis

GeoCorrect will test whether passive evidence can resolve disputed provider locations and whether its conclusions persist over time. For the full-day study, the pipeline will process every IPv4 address observed in a frozen window of the continuous, uniformly random one-percent HTTP-request sample across 765 operational points of presence in 337 cities. The longitudinal study will apply the same evidence definitions and classification pipeline over one month. Both extend the implemented hourly pipeline; the sample is over requests, not one percent of unique IP addresses.

The pipeline will first assemble the provider claim, available RTT observations, and any customer correction report from the relevant time. It will then derive a physical constraint from the RTTs and check the provider location and any proposed alternative against that constraint. An RTT-only correction requires a minimum RTT below the predeclared 2-ms threshold from an operational point of presence. This supports a tightly bounded feasible region rather than an exact coordinate. A customer correction can support an alternative under a less tight RTT bound when the report and physical constraint agree. If the evidence rejects the provider claim but supports no alternative, the pipeline will retain the rejection without forcing a correction.

For each analysis window, the pipeline will assign each observed address one of six mutually exclusive terminal labels. Each label describes the final outcome under the evidence available for that address-window:

- `PROVIDER_PLAUSIBLE`: usable RTT evidence is consistent with the provider location and no applicable evidence contradicts it.
- `PROVIDER_INVALID_RTT`: RTT evidence rejects the provider location but does not support a specific alternative under the correction rules.
- `CORRECTION_SUPPORTED_RTT`: sub-2-ms RTT evidence supports a bounded alternative without a customer report.
- `CORRECTION_SUPPORTED_CUSTOMER_RTT`: a customer correction and the applicable RTT constraint jointly support a bounded alternative.
- `CONFLICTING_EVIDENCE`: the available evidence channels imply incompatible outcomes.
- `INSUFFICIENT_EVIDENCE`: the evidence cannot adjudicate the provider claim or support an alternative.

For every classification, the pipeline will save a versioned manifest: a record of the input window, provider snapshot, RTT evidence, customer evidence when present, and rule version. These records will let us account for all addresses in the full-day sample and determine which outcomes persist during the month-long study. Each result will state the decision supported by that evidence and, when a correction is supported, its feasible region.

11.2 Miscounting: mapping audit and longitudinal concentration

Miscounting will test which concentration trends persist across the identity models introduced in Chapter 7. We will first review candidate ASN groupings to construct a reproducible model containing only accepted merges. We will then compare concentration and diversity over time using dated mappings and routing inputs. RPKI and customer-cone analyses will show how the same identity choices affect related measures.

11.2.1 Operator-mapping audit

The manual mapping audit remains pending. Reviewers will examine candidate sibling clusters—sets of ASNs that an automated mapping associates with one operator. The final mapping will merge only clusters supported by accepted review evidence. It will leave unreviewed, rejected, or ambiguous relationships as singleton ASNs, each forming its own group. This conservative model will record supported merges without claiming a complete operator census.

Before review, we will freeze and enumerate the complete population of candidate clusters. For every candidate, we will record the member ASNs, the automated mappings that proposed the relationship, and dated public evidence. Two reviewers will independently assign an *accept*, *reject*, or *ambiguous* judgment under a written rubric before comparing decisions. They will then adjudicate disagreements using the recorded evidence. The audit-calibrated model will include only clusters accepted by consensus or adjudication; unresolved cases will remain singleton ASNs.

The resulting artifacts will include the frozen candidate list, evidence rubric, pre-adjudication labels, adjudication record, final mapping, and a manifest identifying the routing inputs and analysis code used with the mapping. We will use these records to regenerate the concentration, diversity, topology, RPKI, MOAS, and churn analyses. Chapter 12 specifies the reviewer-agreement and reproducibility checks. The accepted merges address false independence; they do not resolve the distinct service actors concealed by a network identity.

11.2.2 Longitudinal concentration and supporting sensitivity analyses

We will analyze the documented 24-month panel from May 2024 through April 2026. To avoid interpreting the past with later information, we will pair each routing snapshot with the nearest prior organizational mapping. For the RPKI analysis, we will preserve the exact alignment between Prefix2AS and validated ROA payload dates. A route origin authorization applies to an ASN. Grouping sibling ASNs will let us compare operator-level interpretations without changing that protocol fact.

The analysis will compare monthly concentration and diversity under the ASN baseline, the automated mappings, agreement and union scenarios, and the accepted audit model. Each series will retain its effective metric denominator and mapped and unmapped populations. Holding routing inputs fixed does not by itself guarantee that mapping coverage leaves identical effective populations; coverage differences will remain visible in the comparison.

The RPKI analysis will regenerate valid, invalid, and not-found populations for every month and report sibling heterogeneity with exact route-origin and operator-month denominators. It will distinguish whether a validated ROA payload (VRP) exists, whether a route origin passes validation, and how we summarize these results for an organization. These measures answer different questions about security posture.

For customer cones, the analysis will take the union of downstream ASNs in the sibling-ASN cones, counting each shared descendant once. It will separately report edges whose endpoints fall within the same group under an identity model. We will manually review the twenty largest operator-level changes to identify overlap, missing siblings, mapping artifacts, and sensitivity to cone construction. The resulting package will record the aligned inputs and the range of results across models. If a preliminary effect weakens or disappears after these checks, we will report the revised sensitivity result.

11.3 Resolver-Path: deployment evaluation

The remaining work asks whether selected deployed DNS services pass cooperative reports through to their recipients and whether clients can continue checking those reports as keys change or become unavailable. These experiments extend the controlled testbeds to deployment conditions.

11.3.1 Public-Internet option survival

To measure option survival—whether the experimental EDNS option reaches its intended recipient—we will query the major public recursive services represented in the measurement corpus. The experiment will run from at least three independently routed regions over seven days, using randomized names below a controlled authoritative zone.

From each vantage, the experiment will first send ordinary DNS control queries to each resolver. We will interpret option survival for a resolver–vantage pair only after it produces at least 100 successful baseline queries. The experiment will then send matched queries carrying the option. Authoritative logs and client responses will distinguish request-option arrival, response-option return, stripping, modification, ordinary query failure, and TCP fallback. Every attempted resolver–vantage pair will remain in the accounting, including pairs that fail the baseline support threshold. The study will use documented public services and cooperating infrastructure; it will not scan arbitrary open resolvers.

The experiment will produce a resolver–vantage manifest, raw request and response observations, and separate request- and response-survival estimates. Low or zero survival would bound incremental deployment and support a design that requires explicit participation or negotiation rather than legacy passthrough.

11.3.2 Key lifecycle and external deployment

Clients need to retrieve an authorized signing key and respond correctly when that key changes or becomes unusable. The completed prototype generates and validates the Resolver-Path RPKI ASN-to-key object under a private test trust anchor. We will next test publication and retrieval through a production-like repository, rollover, emergency revocation, and unavailable or stale objects. We will also test production-scale retention of nonces, fresh client-generated values used to detect replay. If the dissertation schedule does not allow allocation and standardization of a production object type, we will use a documented experimental identifier and state that the results do not demonstrate standardization or production publication.

We will also attempt a Resolver-Path deployment under independent administration and retain its configuration, implementation provenance, logs, packet captures, and outcome. A broader matrix will test any additional available resolver stacks and operational cache behavior. These experiments extend the completed BIND, Unbound, and Knot testbeds and the separate Go and Python specification-vector implementations.

Dissertation integration and provenance. Aleph, Bet, GOVT-MX, and the core DNSTRACING measurement and system work provide completed evidence for their respective chapters. Bet's annotated 30-AS comparison, naming-class coherence and splitting diagnostics, extraction and coverage analyses, and hyperparameter sensitivity are completed studies. Its remaining work archives the run, scoring, and uncertainty-analysis provenance and integrates those results into the dissertation. The completed DNSTRACING private-ingress analysis, response-path and cache validation, wire and transport tests, response-bound attestation, cross-implementation vectors, and signed performance tests likewise remain completed evidence. During chapter integration, we will align claims, populations, terminology, and artifact provenance across these completed studies.

Chapter 12

Evaluation plan

The evaluation will determine whether each study in Chapter 11 produces reproducible conclusions and handles missing, conflicting, or invalid evidence as specified. We will check internal consistency and the required state transitions before interpreting the empirical results. Reviewer agreement, GeoCorrect label frequencies, sensitivity to identity models, option survival, and deployment availability are outcomes to measure; none must reach a favorable value for completion.

12.1 GeoCorrect: accounting, constraints, and persistence

We will first check that the pipeline assigns exactly one of the six terminal labels to every IPv4 address in each analysis window of the frozen one-percent sample. The report will account for requests and distinct addresses, the coverage of provider snapshots, RTTs, and customer reports, and every exclusion or record that we could not join. Another researcher must be able to reproduce each label from the saved provider claim, RTT constraint, customer correction when present, rule version, and observation time.

Automated checks will confirm that the classification rules cannot assign multiple labels to the same address-window. They will also check that the pipeline applies the sub-2-ms requirement only to the RTT-only correction class and requires customer-supported corrections to agree with the applicable RTT constraint. If RTTs reject a provider location without supporting an alternative, and no evidence conflicts with that rejection, the pipeline must retain `PROVIDER_INVALID_RTT`; if evidence channels conflict, it must record that conflict. We will also conduct a stratified manual audit covering all six labels and retain the evidence used to adjudicate cases. This audit will test whether the implementation follows the rules, without treating reviewers' judgments as ground truth for each address's physical city.

The other evaluation populations introduced in Chapter 6 will test the consequences of these decisions. Independently located stable controls will measure erroneous exclusion within their covered population. Customer disputes and linked policies will establish the reported requirement and whether exclusion would settle it. Neither control error rates nor complaint frequencies will serve as an estimate for all observed addresses.

For the month-long evaluation, we will keep the pipeline and rule version fixed so that changes in outcomes reflect the observations available to that analysis. We will report coverage, label distributions, which outcomes persist, transitions between labels, and sensitivity to evidence availability. Addresses seen on only one day or lacking usable RTT or customer evidence will remain in the denominator; we will not count them as stable corrections. Completion requires full accounting and reproducible application of the evidence rules, without requiring a minimum correction rate, agreement among all evidence channels, or universal city-level ground truth.

12.2 Miscounting: mapping audit and longitudinal comparisons

12.2.1 Mapping-audit reproducibility and agreement

The mapping audit will measure both the support for each accepted merge and the consistency of reviewers' judgments. Every candidate in the frozen population must have two independent labels recorded before adjudication, along with the supporting evidence. We will describe the population by source, cluster size, and routed-address-mass stratum. We will report raw reviewer agreement, Cohen's κ [68], disagreement counts and types, and the disposition of ambiguous cases. Measuring agreement before adjudication preserves the differences in how reviewers interpret the evidence.

For every merge in the audit-calibrated model, we will require a record of acceptance by consensus or adjudication. The model will leave disputed or unresolved relationships as singleton ASNs. We will also regenerate the final mapping and each downstream table from the frozen manifest in a clean environment. If reviewers agree on few relationships, we will retain the resulting lower coverage and report the difficulty of recovering operational groups; the audit will not force consensus to increase coverage.

12.2.2 Historical identity-sensitivity validation

To make the historical comparisons reproducible, we will publish exact dates and populations for routing observations, mappings, VRPs, relationship edges, and customer-cone snapshots. Automated checks will verify that each historical join uses the nearest prior identity mapping and that Prefix2AS and RPKI dates match as specified. They will also confirm that grouping preserves route-origin totals and that each operator's combined customer cone counts a shared descendant only once.

For every month and identity model, we will report the population and denominator used to calculate concentration and diversity, including unmatched routing observations and losses from incomplete mapping coverage. We will distinguish comparisons on a common eligible population from summaries of each model's coverage. Preserving the routing input totals is an accounting check; it does not establish that all previously reported concentration estimates used identical denominators.

Results will report an ASN baseline, each automated organization mapping, the conservative-agreement and loose-union scenarios, and the accepted audit scenario. We will separate static from longitudinal analyses. For RPKI, we will retain the valid, invalid, and not-found route-origin denominators and distinguish authorization at the ASN level from differences among sibling ASNs. For customer cones, we will report how we combined cones, which edges became internal to a group, the range across models, and the manual findings for the twenty largest operator-level changes.

The study is complete when the analysis uses aligned inputs, passes these accounting checks, and shows which conclusions remain stable across identity models. The preliminary concentration, RPKI, churn, or cone effect sizes need not persist.

12.3 Resolver-Path: option survival and operational verification

12.3.1 Public-Internet option-survival measurement

The experiment will attempt queries to every selected public resolver from every planned vantage. For each resolver–vantage pair, we will record baseline success, request-option arrival, response-option return, stripping or modification, timeout, and TCP fallback in the evaluation manifest. We will mark a pair with fewer than 100 successful baseline queries as insufficiently supported and retain it in the accounting.

We will report request and response survival separately, with confidence intervals and per-resolver, per-vantage, and per-day breakdowns. Ordinary-query controls will distinguish general reachability failure from option-specific failure. Completion requires measuring and bounding survival over the declared population, without requiring a favorable survival percentage.

12.3.2 Production key and external-deployment validation

To check whether clients handle changing key availability correctly, the production-like Resolver-Path experiment will publish, retrieve, and verify the RPKI ASN-to-key object. We will record outcomes for rollover, revocation, stale-object, missing-object, invalid-resource, and unavailable-repository cases. In each case, the verifier must produce the specified valid, invalid, unattested, or unknown/revoked-key state without treating RPKI authorization as proof of the country field. The nonce-state experiment will report memory, lookup cost, expiration behavior, and duplicate-response handling at declared concurrency and query rates.

We will evaluate an independently administered deployment when a partner and operational approval are available. We will record the resolver implementation, administrative boundary, configuration, path scenario, ordinary-query control, option result, logs, and packet capture. The evaluation matrix will retain cases involving additional resolver stacks and operational cache behavior even when a case fails. If no partner is available, a documented attempt plus the completed production-like key-lifecycle matrix satisfies the planned task; it does not substitute for external-deployment evidence.

Chapter 10 reports the completed response-bound signing evaluation: 18 canonical vectors in two standalone codebases, the wire and TCP model across 48 conditions, and the 450-run signed depth/load matrix. Those controlled results establish the baseline for the new deployment evaluation; they do not substitute for external or production evidence.

Completion requires reproducible evidence and correct, bounded behavior. Publication, persistent effect sizes, favorable survival, standardization, and partner deployment are not prerequisites.

Chapter 13

Expected contributions

The dissertation will contribute methods, empirical findings, and reusable artifacts for three practical questions: where infrastructure may be located, which actors should be distinguished, and what we can learn about request participants from endpoint observations and authenticated reports. The completed studies already answer parts of these questions. The remaining studies will test how those answers change with evidence availability, identity assumptions, and deployment conditions. Throughout, each result will state what the evidence supports and which alternatives remain unresolved.

13.1 Methods for answering bounded infrastructure questions

Aleph and Bet turn operator-local names into reusable extractors and candidate locations. Their benchmark measures recovery of annotated geographic hints; semantic review tests hint meaning, and RTT measurements test physical plausibility. GeoCorrect extends the location work to disputed provider claims. Its proposed studies will determine when passive evidence can reject a claim without identifying a replacement and when stronger evidence supports a bounded correction.

Miscounting makes the ASN-to-operator mapping an explicit assumption of concentration and longitudinal analyses, so researchers can test which conclusions depend on their grouping choices. GOVT-MX attributes inbound handling, hosting, and authorized sending separately, preserving the service actors that a single network identity can conceal. For DNS, endpoint measurement motivates two cooperative disclosure modes and response-bound attestation. This mechanism authenticates received reports while leaving silent participants outside its guarantee.

Across these methods, we record the original observation, its time and population, the target property, the interpretation rule, and the check that supports the conclusion. A researcher can use these records to inspect, repeat, or revise an inference without treating a source as authoritative about properties it cannot establish.

13.2 Internet-scale empirical findings

The completed location studies establish the coverage and limits of reusable PTR-derived city labels at Internet scale. Bet also compares naming-class coherence, extraction, and finest-granularity hint recovery with Aleph on an annotated 30-AS benchmark. These results describe recovery separately from semantic correctness and physical plausibility. GOVT-MX establishes the prevalence of visible inbound provider roles and separately authorized outbound mail roles in its curated public-sector domain corpus.

GeoCorrect will account for every address in the declared one-percent sample for a full day and over a month. It will report how often provider locations are plausible, RTT-invalid, supported by one of the two correction paths, conflicting, or insufficiently evidenced. The longitudinal analysis will show which outcomes persist and which change. These frequencies describe audit outcomes under the available evidence, without estimating Internet-wide city accuracy.

The remaining Miscounting study will show how concentration, diversity, RPKI posture, topology, customer cones, MOAS interpretation, and origin churn vary under documented identity models.

DNSTRACING has established a conservative 6,622-of-16,636 (39.8%) client–frontend AS-boundary lower bound, a separate 6,480-of-9,877 (65.6%) public-ingress result, and bounded country, latency, concentration, and longitudinal findings. Its remaining experiments will measure option survival through selected deployed resolution paths and test how key lifecycle and partial deployment affect received reports. The contribution will include the resulting estimates and sensitivity ranges, even if a preliminary Miscounting effect weakens or option survival is low.

13.3 Systems, datasets, and research artifacts

The dissertation will assemble artifacts that let other researchers inspect and reproduce the claims. The location artifacts include PTR rules, hint cards, benchmark annotations, scoring records, and GeoCorrect manifests recording provider claims, RTTs, customer evidence, and classifications. The identity artifacts include ASN-to-entity mappings, audit records, and mail-role outputs. The DNS artifacts include resolver manifests; BIND, Unbound, and Knot tracing and attestation implementations; response-bound vectors; standalone Go and Python conformance implementations; and linked logs and packet captures. For each result, the reproducibility package will identify the input dates, configuration, code version, and supported claim.

When licensed or sensitive inputs restrict release, we will document what is unavailable and provide derived manifests, substitutes, or replayable interfaces where permitted. Completion requires inspectable methods, evidence, limitations, and reproducible analyses within those data constraints. Publication acceptance, high option survival, and partner deployment remain outcomes rather than completion requirements.

Chapter 14

Work plan and schedule

The schedule records the prospectus's current target milestones. Research and chapter writing will proceed together, with each analysis depending on specified inputs: GeoCorrect requires fixed rules and observation windows; Miscounting requires audited mappings and dated routing inputs; and Resolver-Path's deployment evaluation builds on its completed controlled experiments.

14.1 Remaining milestones

Table 14.1: Planned milestones.

Target date	Milestone
13 November 2026	Prospectus.
December 2026	GOVT-MX resubmission to CoNEXT.
December 2026–February 2027	DNSTRACING and Miscounting validation.
March 2027	Dissertation submission.

The work plan also includes GeoCorrect's full-day and month-long analyses, Bet provenance packaging, and cross-chapter integration.

14.2 Risks and contingencies

Sparse or conflicting GeoCorrect evidence. Some addresses may lack usable RTT observations, appear on only one day, or have customer reports that conflict with RTT constraints. The pipeline will retain these addresses in the declared population and use `INSUFFICIENT_EVIDENCE` or `CONFLICTING_EVIDENCE` when the classification rules require them. The longitudinal analysis will not treat a single day's observation as a stable correction. We will keep the correction rule fixed even if few addresses meet it. If the sub-2-ms rule supports only a region broader than a city, the reported correction will retain that region.

Mapping disagreement. Reviewers may disagree about candidate sibling relationships. The audit-calibrated model will include only relationships accepted by consensus or explicit adjudication. If disagreement reduces accepted coverage, we will report that result without relaxing the evidence standard.

Historical-data gaps. If we cannot recover or align an input snapshot, we will identify and exclude the affected month under a documented rule. We will narrow historical claims to the resulting common panel and will not use later mappings to fill a missing period.

Low option survival. Even if most deployed resolvers strip the experimental option, the measurement will establish where survival fails. We will use that result to identify where the mechanism requires explicit participation or negotiation rather than relying on transparent passthrough.

External deployment. An independently administered deployment depends on a partner and operational approval. If no partner deployment is available within the evaluation window, we will document the attempt and complete the expanded controlled matrix, including partial deployment and failure cases, and the production-like key-lifecycle matrix. This satisfies the task defined in Chapter 12, while leaving external deployment untested. External deployment is best-effort evidence, not a prerequisite for the defense.

Key-binding standardization. The implemented Resolver-Path object validates an ASN-to-key binding under a private RPKI trust anchor. Its object type and production publication are not standardized. If standards allocation remains unavailable on the dissertation schedule, we will use a documented experimental identifier and state this limitation. Operating an Internet-wide public-key infrastructure or completing standardization is outside the dissertation.

14.3 Dissertation integration and submission

We will integrate completed chapters as the remaining evidence stabilizes. For Bet, we will archive run, scorer, learning-input, and uncertainty-analysis provenance alongside the completed results. For GeoCorrect and Miscounting, we will assemble reproducible full-study outputs. For DNSTRACING, we will incorporate option-survival, key-lifecycle, and deployment findings alongside the completed controlled measurement and system evaluation.

The synthesis will distinguish semantic correctness, RTT compatibility, identity assumptions, role attribution, and report verification. Artifact manifests will record each result's inputs, observation dates, code, configuration, and supported claims. Committee feedback, figure and reference checks, and artifact preparation precede the March 2027 submission target.

Chapter 15

Conclusion

Partial evidence can support bounded conclusions about infrastructure location, operational identity, and request participation when the interpretation from observation to claim is explicit and the resulting claim is tested against appropriate evidence. A useful answer need not identify an exact site, enumerate every operator, or expose every participant. It must explain how the observations support the conclusion and which alternatives remain possible.

Aleph and Bet demonstrate scalable geographic-hint recovery, with semantic review and RTT checks addressing meaning and physical plausibility separately. Miscounting's preliminary sensitivity and churn findings motivate the remaining mapping audit and longitudinal analysis. GOVT-MX distinguishes hosting, inbound handling, and authorized sending. DNS endpoint measurements reveal why additional disclosure is needed; cooperative reporting and response-bound attestation let recipients authenticate received reports while leaving silent participants outside the guarantee.

The remaining studies will evaluate GeoCorrect's passive evidence over full-day and month-long windows, test concentration under audited identity mappings, and examine Resolver-Path under deployment conditions. Reproducible answers with explicit populations and evidentiary limits will complete this work. Together, the studies provide methods for learning from incomplete visibility and making unresolved alternatives inspectable.

Acknowledgments. We conduct the Bet work in association with IPinfo and GeoCorrect in association with Cloudflare.

Appendix A

Regulatory declarations as an application of bounded evidence

This appendix shows how a regulatory auditor could use the dissertation's methods to investigate a specific declaration about an ICT service. Table A.1 connects DORA declarations to technical checks and identifies who could act on a discrepancy [3]. An observation that conflicts with a declaration warrants investigation; it does not establish non-compliance. Agreement likewise does not establish that the declaration is complete.

Before applying any check, the auditor needs access to the financial entity's register, a clearly scoped ICT service, and evidence linking that service to its Internet identifiers at the relevant time. The current studies do not have these inputs. The mapping below therefore describes a possible future application.

Commission Implementing Regulation (EU) 2024/2956 structures the Article 28 register into templates [69]. The fields relevant to the technical checks describe locations, providers, supply chains, and substitutability:

- Template `B_02.02` records the country of service provision (`B_02.02.0130`), data-at-rest location (`B_02.02.0150`), and processing location (`B_02.02.0160`). These fields concern different location claims.
- Template `B_05.01` identifies each ICT third-party provider. For legal persons, it uses a legal entity identifier (LEI) or European unique identifier (EUID) (`B_05.01.0010-.0020`). It also records the provider's ultimate parent (`B_05.01.0110-.0120`).
- Template `B_05.02` links a contractual arrangement (`.0010`), ICT service type (`.0020`), provider identifier (`.0030`), supply-chain rank (`.0050`), and recipient of the subcontracted service (`.0060`).
- Template `B_07.01.0050` records the financial entity's service-specific assessment of substitutability.

Commission Delegated Regulation (EU) 2025/532 additionally requires assessment of the location and complexity of subcontracting chains and identification of subcontractors supporting critical or important functions or material parts of them [70]. These declarations provide the starting point for the conditional checks in the table.

An auditor would carry out each comparison in five steps:

1. Select a register declaration and its effective time.
2. Link the contracted service to its technical identifiers at that time.
3. Collect evidence that tests the property named in the declaration.
4. Compare the observation with the declared value.
5. Refer a discrepancy to a party that can inspect contracts, internal telemetry, and provider explanations.

Aleph, Bet, and Miscounting can support the location and identity checks once an auditor links the service to its technical identifiers. GOVT-MX demonstrates role-specific attribution on public-sector mail domains; applying it here would require a corpus of DORA-scoped financial domains. DNSTRACING demonstrates endpoint measurement and cooperative reporting, but a deployment would also need to link signer identities to registered providers.

Table A.1: DORA declarations, the technical checks they could support, and the limits of those checks. Register fields are defined by Regulation (EU) 2024/2956 and subcontracting duties by Regulation (EU) 2025/532 [69, 70].

Audit object	**Declaration**	**Technical check and contrary evidence**	**Actor that can act**	**Not established**
Location (Aleph/Bet)	`B_02.02.0130`: country of ICT-service provision. `.0150-.0160`: storage and processing countries.	For a registered service bound to a PTR-named endpoint, a PTR-derived country outside all declared service-provision countries that survives operator review or a separate latency-plausibility check would trigger review of `.0130`.	Financial entity's service owner, third-party-risk function, or competent authority.	Contracted-function location, storage or processing location, data residency, and jurisdiction; Aleph and Bet cannot test `.0150-.0160`.
Provider identity and concentration (Miscounting)	`B_05.01.0010-.0020`: provider LEI/EUID; `.0110-.0120`: ultimate parent; `B_07.01.0050`: service-specific substitutability.	For registered services with observed origin ASNs, repeated collapse of nominally distinct technical dependencies into one candidate operational group, or one provider split across ASNs, would challenge a network-diversity analysis.	Financial entity's third-party-risk and concentration-risk functions or competent authority.	Legal identity, close connection, substitutability, common control, and correlated failure; ASN mappings do not contradict an LEI/EUID record.
Service roles and delegation (GOVT-MX)	`B_05.02.0010-.0070`: contractual reference, ICT-service type, provider identifier, supply-chain rank, and recipient of subcontracted service; Regulation 2025/532: relevant subcontractor identification.	For a registered email service, a provider-specific inbound role or domain-authorized outbound role absent from the corresponding declared chain would warrant provider and contract review. GOVT-MX would require a DORA-scoped financial-domain corpus to perform this check.	Financial entity's service and contract owners, direct provider, or competent authority.	Contractual rank, actual use of an SPF-authorized sender, mailbox or data location, hidden processing, legal responsibility, and register completeness.
Request participation (DNSTRACING)	Article 8(5): processes dependent on ICT third parties and their interconnections; `B_05.02.0010-.0070`: declared ICT service supply chain.	For a registered DNS service, a cryptographically authenticated resolver participant absent from the declared chain would warrant dependency review. This requires cooperative deployment and a binding from signer identity to a registered provider.	Financial entity's DNS or service owner, direct provider, or competent authority.	Silent or omitted participants, a complete request path, legal provider identity without key binding, service criticality, and contractual responsibility.

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